

Core-Collapse Supernovae from Single and Binary-Stripped Massive Stars – I. Explodability and Nucleosynthesis

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ABSTRACT

Binary interaction can substantially alter massive-star evolution and, in turn, pre-supernova structure, potentially changing core-collapse outcomes and nucleosynthetic yields. We systematically compare explodability and nucleosynthesis in non-rotating, solar-metallicity single-star and binary-stripped progenitors spanning zero-age main-sequence (ZAMS) masses of 11–45 $M_{\odot}$, using the parameterised, spherically symmetric PUSH explosion framework and detailed nucleosynthesis post-processing. Binary stripping alters the explodability landscape without systematically favouring successful or failed explosions. Successfully exploding models from both populations occupy similar regions of explosion parameter space, with explosion energies of ~ 0.65 – 1.3 B and neutron star remnant masses of ~ 1.5 – 2.0 $M_{\odot}$. The largest nucleosynthetic differences occur at low ZAMS masses, particularly for material undergoing explosive O and Si burning. The predicted ^{56}Ni yields and iron-group abundances are broadly consistent with observations of core-collapse supernovae and the metal-poor star HD 84937. We find that the Salpeter initial-mass-function-weighted yields of most elements differ by only $\sim 20\%$ between the two populations, with K showing the largest difference with an enhancement factor of ~ 3.4 in the binary-stripped population. Notably, this enhancement reduces the underproduction of K relative to Mg compared with empirical core-collapse abundance ratios inferred from APOGEE, while Na remains overproduced in both populations. Although the adopted explosion treatment affects the predicted explodability landscape, binarity influences individual outcomes and specific elemental yields, with consequences for Galactic chemical evolution.

Key words: Nucleosynthesis – Transients: supernovae – Methods: numerical – binaries: general – stars: massive

1 INTRODUCTION

Core-collapse supernovae (CCSNe) mark the energetic deaths of massive stars. At lower zero-age main-sequence masses ($M_{\text{ZAMS}} \simeq 8$ – $10 M_{\odot}$), they may arise from the collapse of an ONeMg core, while more massive stars undergo collapse after forming an Fe core. During these explosions, newly synthesised material is ejected into the surrounding environment, enriching the interstellar medium with intermediate-mass and iron-group elements. CCSNe therefore provide an important source of chemical enrichment and feedback, playing a central role in the chemical evolution of galaxies.

Most massive stars do not evolve in complete isolation. A large fraction are found in multiple-star systems (Abt 1983; Mason et al.

2009), and many are expected to undergo at least one phase of mass transfer during their lifetimes (Sana et al. 2012; Moe & Di Stefano 2017). Such binary interactions can fundamentally alter their evolutionary pathways (Podsiadlowski et al. 1992; Brown et al. 1996; Wellstein & Langer 1999; Eldridge et al. 2008; Langer 2012; de Mink et al. 2013). During mass transfer, much of the primary star's outer hydrogen-rich envelope may be removed, with some fraction of this material accreted by the companion. This alters the subsequent stellar evolution and final pre-collapse structure, with consequences for the chemical yields of the system (Braun & Langer 1995b; Laplace et al. 2021; Farmer et al. 2021, 2023).

Despite the importance of binarity, explosive nucleosynthesis in binary-stripped progenitors has been less extensively explored (Braun & Langer 1995a; Vartanyan et al. 2021; Farmer et al. 2023; Boccioli & Roberti 2026; Jalin et al. 2026) than for single-star pro-

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genitors (e.g., Woosley et al. 1993; Heger et al. 2001, 2002; Heger & Woosley 2002; Woosley & Heger 2007; Pignatari et al. 2016; Limongi & Chieffi 2018; Curtis et al. 2019; Ebinger et al. 2020a; Andrews et al. 2020; Sieverding et al. 2023). A central limitation has been the availability of detailed binary progenitor models evolved through the final stages of burning to the point of core collapse. Evolving binary progenitors through these final stages requires detailed nuclear reaction networks, as electron capture reactions can alter the structure and composition, while even relatively small differences in the pre-collapse structure can affect the subsequent outcome (Farmer et al. 2016; Vartanyan et al. 2018; Laplace et al. 2021). The computational cost of these larger reaction networks has therefore made it difficult to produce extensive grids of binary progenitors evolved to core collapse (Laplace et al. 2021).

The difficulty of modelling CCSNe does not end with producing suitably detailed pre-collapse progenitor grids, as the explosion mechanism itself also remains challenging to model. It requires the coupled treatment of hydrodynamics, gravity, nuclear equations of state, neutrino transport, nuclear reaction rates, and possibly neutrino flavour conversion. Although major progress has been made in multidimensional simulations over the last decade, fully self-consistent calculations remain computationally expensive and are not yet practical for population-level studies. Although a growing number of multidimensional simulations now follow the evolution from collapse to shock revival (e.g. Janka 2012; Burrows 2013; Nakamura et al. 2015; Janka et al. 2016; Bruenn et al. 2016b; Burrows et al. 2018, 2020; Vartanyan & Burrows 2023; Sandoval et al. 2021; Burrows et al. 2024, 2026; Neopane et al. 2026; Jalin et al. 2026), uncertainties remain in which progenitors explode, what explosion energies they produce, and how these results vary across large progenitor grids (e.g. Takiwaki et al. 2014; Lentz et al. 2015; Melson et al. 2015a,b; Müller 2015; Janka et al. 2016; Hix et al. 2016). For a recent review of advances in multidimensional CCSN modelling, see Janka (2025).

In one-dimensional (1D), self-consistent, spherically symmetric simulations, CCSNe generally do not explode through the delayed neutrino-driven mechanism alone, except at the low-mass end of the progenitor population (e.g. Liebendörfer et al. 2004). Successful shock revival for more massive progenitors appears to require multidimensional hydrodynamic effects, particularly neutrino-driven convection and accretion shock instability, which increase the efficiency of neutrino heating behind the stalled shock (e.g. Herant et al. 1994; Janka & Mueller 1996; Blondin et al. 2003; Foglizzo et al. 2007). Alternatively, rapid rotation and magnetic-field amplification may produce magnetorotational explosions in a subset of strongly magnetised progenitors (Winteler et al. 2012; Mösta et al. 2014, 2015; Nishimura et al. 2015, 2017; Halevi & Mösta 2018; Kuroda et al. 2020; Obergaulinger & Aloy 2020; Aloy & Obergaulinger 2021).

As multidimensional simulations remain computationally expensive, several 1D approaches have been developed to parametrise the effects that facilitate successful explosions and apply them systematically across large progenitor grids. Simple prescriptions such as piston and thermal-bomb explosions, in which the shock is launched by moving an inner mass shell or by depositing thermal energy into the innermost stellar layers, have been widely used for nucleosynthesis calculations (e.g. Aufderheide et al. 1991; Woosley & Weaver 1995; Sawada & Maeda 2019). These approaches remain in use in recent studies (e.g., Farmer et al. 2023), and provide an efficient way to calculate nucleosynthetic yields across large progenitor grids. However, key explosion properties (i.e., the explosion energy and mass cut) are imposed rather than emerging from the explosion dynamics, limiting their ability to predict the nucleosynthesis of the innermost ejecta (e.g. Thielemann et al. 1996; Woosley & Weaver 1995; Rauscher

et al. 2002; Limongi & Chieffi 2006; Woosley & Heger 2007; Umeda & Nomoto 2008; Limongi & Chieffi 2012). More physically motivated 1D approaches attempt to approximate the multidimensional effects that facilitate shock revival. The Supernova Turbulence In Reduced-dimensionality (STIR) model incorporates neutrino-driven convection and turbulence through a time-dependent mixing-length prescription (Couch et al. 2020; Boccioli et al. 2021; Boccioli & Roberti 2026). The PUSH method, which is used in this study, instead introduces an additional source of neutrino heating to facilitate shock revival while retaining detailed neutrino transport (e.g., Perego et al. 2015; Ebinger et al. 2019). These approaches provide a framework for systematically studying explosion outcomes, remnant properties, and yields across large progenitor grids.

Assessing the impact of different stellar-evolution pathways requires considering not only how they alter the explodability of the resulting progenitors, but also how they affect the composition of the ejecta and the resulting observables (e.g. Nomoto et al. 2013; Bruenn et al. 2016b; Caffau et al. 2013). Nucleosynthetic yields therefore provide an important link between the pre-supernova evolution, the explosion dynamics, and the final composition of the ejecta. These yields depend on both the pre-supernova composition and stratification of the progenitor and on which regions ultimately become unbound during the explosion. In the outer ejecta, the final abundances are largely inherited from hydrostatic burning during stellar evolution, with the explosion primarily determining whether these layers are successfully ejected. This material contains the products of H, He, C, Ne, and O burning, including He, C, O, Ne, Mg, and other intermediate-mass elements. In contrast, the composition of the innermost ejecta is considerably more sensitive to the explosion itself, with the production of iron-group elements depending on the shock strength, the thermodynamic histories of the ejecta, the location of the mass cut, and the electron fraction, Y_e (Curtis et al. 2019). Binary stripping can therefore influence the nucleosynthetic yields both by altering the pre-collapse composition and stratification of the progenitor and, through these structural differences, by affecting the conditions of the subsequent explosion.

In this work, we use the PUSH framework to simulate the single-star and binary-stripped progenitors of Farmer et al. (2023), allowing us to compare their explosion properties and nucleosynthetic yields consistently. We investigate how binary stripping alters the explodability of the progenitors, the resulting explosion energies and remnant masses, and the composition of the ejecta. The paper is structured as follows. In Section 2, we describe the progenitor models, the PUSH framework, and the nucleosynthesis post-processing. In Section 3, we show the explosion outcomes and properties across the progenitor grid and examine their dependence on ZAMS mass, compactness, and the adopted PUSH calibration. In Section 4, we explore the nucleosynthetic yields and the differences between the single-star and binary-stripped models. In Section 5, we compare our predictions with observational constraints from CCSNe and metal-poor stars, and with empirical core-collapse abundance ratios inferred from APOGEE stellar abundances. Finally, we summarise our main findings and conclusions in Section 6.

2 METHOD AND INPUTS

2.1 Input Models

In this study, we consider the two progenitor sets generated by Farmer et al. (2023): isolated single stars (the “Ms” series) and binary-stripped stars that have undergone Roche-lobe overflow (the “Mb”

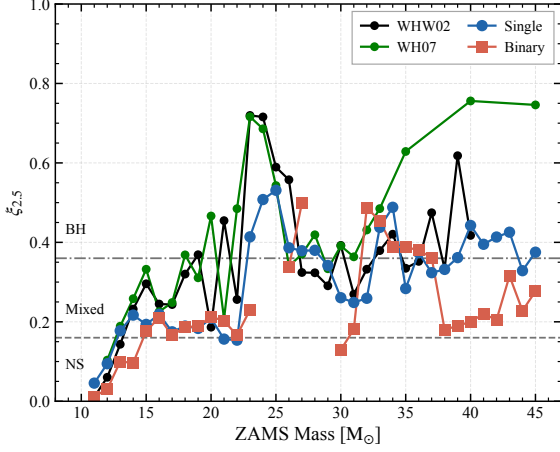


Figure 1. Compactness at $2.5 M_{\odot}$ ($\xi_{2.5}$) across the progenitor ZAMS mass range for single and binary models. The horizontal thresholds at $\xi_{2.5} = 0.16$ and 0.36 indicate the approximate transition between neutron star, mixed, and black hole outcomes (Farmer et al. 2023). Solar-metallicity progenitors from (Woosley et al. 2002; Woosley & Heger 2007) are overplotted for comparison.

series). Both sets were calculated with the MESA stellar evolution code (Paxton et al. 2011, 2013, 2015, 2018, 2019; Jermyn et al. 2023). Importantly, the Mb progenitors differ from their single-star counterparts not only in their envelope masses, but also in their core masses, composition gradients, and subsequent shell-burning evolution.

All progenitors are non-rotating and assume a solar metallicity of $Z = 0.0142$. The single-star and binary-stripped models adopt an initial helium mass fraction of $Y = 0.2684$ ($Y = 2Z + 0.24$; Pols et al. 1995; Tout et al. 1996), with the initial abundance pattern based on the solar composition of Grevesse & Sauval (1998). The progenitor grids span ZAMS masses of 11–45 $M_{\odot}$ in intervals of 1 $M_{\odot}$. Four Mb models, with ZAMS masses of 24, 25, 28, and 29 $M_{\odot}$, are not evolved to the point of core-collapse within MESA due to numerical issues and are therefore excluded from our analysis. Throughout this work, we place our results in the context of previous solar-metallicity PUSH calculations using the S-series (Woosley et al. 2002) and W-series (Woosley & Heger 2007) progenitors presented by Ebinger et al. (2019); Curtis et al. (2019) which are specified as WHW02 and WH07 throughout. Note that these single-star models were calculated from the KEPLER code, presented in Woosley et al. (2002) and Woosley & Heger (2007).

Following previous studies, we characterise the pre-collapse core structure using the compactness parameter introduced by O’Connor & Ott (2011). For consistency with the explodability analysis of Farmer et al. (2023), we evaluate the compactness at an enclosed mass of $2.5 M_{\odot}$:

$$\xi_{2.5} = \frac{2.5}{R(M = 2.5 M_{\odot})/1000 \text{ km}}, \quad (1)$$

where $R(M = 2.5 M_{\odot})$ is the radius enclosing a baryonic mass of $2.5 M_{\odot}$. Figure 1 shows $\xi_{2.5}$ as a function of ZAMS mass for the Ms, Mb, WHW02 and WH07 progenitor series. The compactness varies non-linearly with ZAMS mass, reflecting the sensitivity of the final core structure to shell-burning evolution, semiconvection, convective overshooting, and mass loss. Note that the PUSH calibration described below is based on the compactness at core bounce, $\xi_{2.0}$, following Ebinger et al. (2019). The corresponding $\xi_{2.0}$ trends with ZAMS mass are shown in Appendix A.

2.2 The PUSH Framework

We adopt the standard PUSH setup used in previous studies to follow the collapse, core bounce, shock revival, and the subsequent post-bounce evolution (see Ebinger et al. 2019; Curtis et al. 2019; Ebinger et al. 2020b). The resulting simulations provide the explosion properties, remnant masses, and thermodynamic trajectories required for nucleosynthesis post-processing. PUSH is based on the delayed neutrino-driven explosion mechanism and introduces an additional source of heating in the gain region of spherically symmetric simulations. This additional heating is motivated by the increased net heating experienced by fluid elements in multidimensional simulations, where convection and accretion can increase the efficiency of neutrino energy deposition. Specifically, PUSH deposits a fraction of the heavy-flavour neutrino energy ($\nu_x = \nu_{\mu}, \bar{\nu}_{\mu}, \nu_{\tau}, \bar{\nu}_{\tau}$) behind the shock to trigger successful explosions.

The method (Perego et al. 2015) employs the general-relativistic hydrodynamics code Agile, coupled to neutrino transport. Electron-flavour neutrinos are treated using the isotropic diffusion source approximation (IDSA; Liebendörfer et al. 2001), while heavy-flavour neutrinos are handled with the advanced spectral leakage (ASL; Perego et al. 2015) scheme. For matter in nuclear statistical equilibrium (NSE), we adopt the finite-temperature HS(DD2) equation of state (Hempel & Schaffner-Bielich 2010a). Outside NSE, the matter is described by an ideal-gas equation of state coupled to an α -network.

The PUSH heating term, $Q_{\text{push}}^+(t, r)$, transfers a fraction of the heavy-flavour neutrino energy to matter behind the shock and depends on the neutrino spectral energy flux and includes both spatial and temporal components, with two free parameters: k_{push} and t_{rise} . We adopt the standard calibration of Ebinger et al. (2019), based on SN 1987A, with $t_{\text{rise}} = 400$ ms and a compactness-dependent scaling

$$k_{\text{push}} = a\xi^2 + b\xi + c, \quad (2)$$

where $a = -23.99$, $b = 13.22$, and $c = 2.5$. Here, $\xi_{2.0}$ denotes the compactness at core bounce.

The simulations include material from the centre to the helium layer and are evolved for 15 s unless black-hole formation occurs earlier. The outcome of each model is classified as: (i) a successful explosion if the explosion energy saturates at a positive value, (ii) black-hole formation if the central density exceeds $\sim 10^{15} \text{ g cm}^{-3}$, or (iii) a failed explosion if the final explosion energy remains negative after PUSH becomes inactive. The explosion energy is computed following Perego et al. (2015) as the sum of thermal, kinetic, and gravitational energy integrated from the mass cut to the surface.

2.3 Nucleosynthesis Post-processing

We perform detailed nucleosynthesis post-processing of successful explosion models using the nuclear reaction network CFNET (Fröhlich et al. 2006). Our setup follows Ebinger et al. (2019); Curtis et al. (2019), tracking 2,902 isotopes, including free nucleons and nuclei on both sides of the valley of β -stability up to ^{211}Eu . Reaction rates are taken from REACLIB (Cyburt et al. 2010), incorporating experimental data where available and theoretical predictions for neutron-, proton-, and α -capture reactions from Rauscher & Thielemann (2000). Weak interactions include electron and positron capture rates from Langanke & Martínez-Pinedo (2001), $\beta^{\pm}$ decays from NuDat¹ and Morrell et al. (2005), and neutrino and antineutrino captures on free nucleons.

¹ <http://www.nndc.bnl.gov/nudat2/>

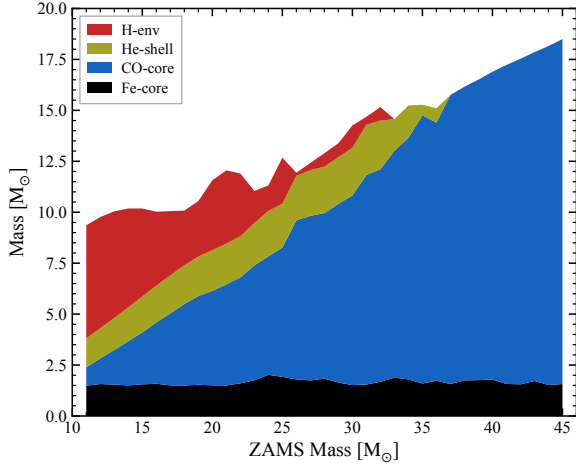


Figure 2. Fe-core mass (black), CO-core mass (blue), He-shell mass (yellow), and total mass including the H-envelope (red) as a function of ZAMS mass at the onset of collapse for the Ms series progenitors (Farmer et al. 2023).

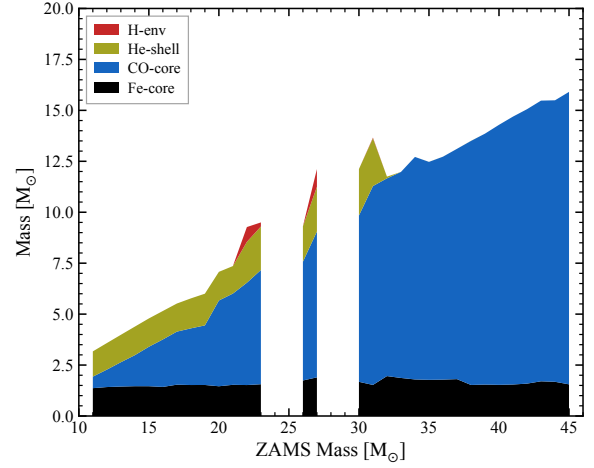


Figure 3. Same as Figure 2, but for the Mb series models (Farmer et al. 2023).

The ejecta are divided into mass elements (“tracers”) of $10^{-3} M_{\odot}$. For each tracer, the thermodynamic history is followed from the onset of collapse to the final hydrodynamic timestep, up to 15 s, or until the shock leaves the computational grid. For tracers exceeding 10 GK, nucleosynthesis calculations begin once the temperature falls below this threshold, assuming NSE abundances as the initial composition. For cooler tracers, the calculations begin at the start of the hydrodynamic evolution, with the initial abundances taken from the nuclear reaction network coupled to the hydrodynamics. Note only tracers reaching temperatures above 1.75 GK are processed. In all cases, the electron fraction is initialised from the hydrodynamic simulation and evolved self-consistently within the network. The tracer trajectories are extrapolated beyond the simulation time by:

$$r(t) = r_{\text{final}} + t v_{\text{final}} \quad (3)$$

$$\rho(t) = \rho_{\text{final}} \left(\frac{t}{t_{\text{final}}} \right)^{-3} \quad (4)$$

$$T(t) = T[s_{\text{final}}, \rho(t), Y_e(t)] \quad (5)$$

where r , v , ρ , T , s , and Y_e denote radius, velocity, density, temperature, entropy per baryon, and electron fraction, respectively. The subscript “final” refers to values at the end of the hydrodynamic simulation. The temperature evolution is obtained using the equation of state of Timmes & Swesty (2000). Nucleosynthesis calculations are terminated once the temperature drops below 0.05 GK.

3 EXPLOSION PROPERTIES ACROSS THE MASS RANGE

In this section, we discuss the explosion outcomes and properties of both progenitor sets, spanning ZAMS masses of 11–45 $M_{\odot}$. Four Mb models (24, 25, 28, and 29 $M_{\odot}$) did not reach core collapse and are therefore excluded. We first examine successful and failed explosions, followed by trends in explosion energies, remnant masses, and explosion times with ZAMS mass and bounce compactness $\xi_{2.0}$.

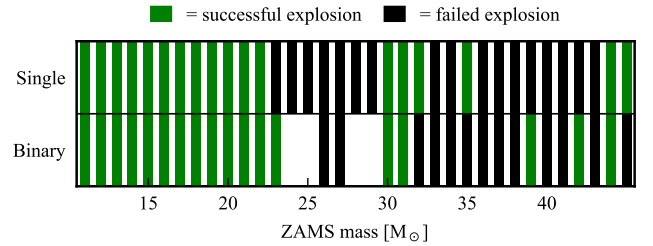


Figure 4. Explosion outcomes as a function of ZAMS mass for the Ms (top) and Mb (bottom) progenitor series, showing successful and failed models; gaps indicate masses for which no Mb model is available.

3.1 Explosion outcomes

Figure 4 summarises the explosion outcomes for the two progenitor series across the full ZAMS-mass range. We find that of the 35 Ms models investigated here 18 explode, corresponding to a success rate of 51.4 per cent, while 18 of the 31 Mb models explode, giving a slightly higher success rate of 58.1 per cent. The two series behave similarly at low ZAMS masses, with all models up to and including 22 $M_{\odot}$ exploding. Above this mass, however, their outcomes begin to diverge: the first Ms failure occurs at 23 $M_{\odot}$, whereas the first Mb failure occurs at 26 $M_{\odot}$. The two series produce opposite outcomes at 23 $M_{\odot}$, 32 $M_{\odot}$, 35 $M_{\odot}$, 42 $M_{\odot}$, and 45 $M_{\odot}$, while both explode at 30 $M_{\odot}$, 31 $M_{\odot}$, and 44 $M_{\odot}$.

Figure 5 compares our results with those of Farmer et al. (2023), using the two-parameter explodability criterion of Ertl et al. (2016), and Boccioli & Roberti (2026), using neutrino-driven explosion simulations with time-dependent mixing-length theory. For the Ms series at $M_{\text{ZAMS}} \leq 22 M_{\odot}$, all of our models explode, while Farmer et al. (2023) predict explosions only at 11, 12, 15, and 21 $M_{\odot}$, and Boccioli & Roberti (2026) at 11, 16, 18, and 20 $M_{\odot}$. Between 23 and 29 $M_{\odot}$, none of our Ms models explode, in agreement with Farmer et al. (2023), whereas Boccioli & Roberti (2026) predict successful explosions for all but one model. At higher ZAMS masses, Boccioli & Roberti (2026) again predict explosions for most models, while ours show more frequent failures and somewhat better agreement

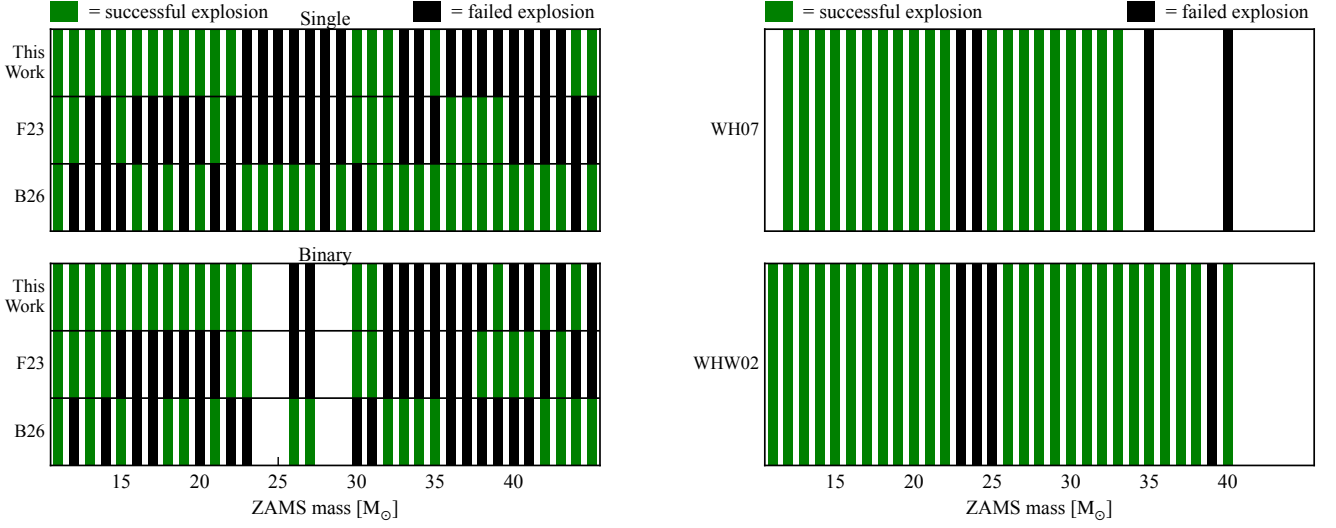


Figure 5. Left: Explosion outcomes as a function of ZAMS mass for the single-star (Ms; top) and binary-stripped (Mb; bottom) progenitor series. The outcomes obtained in this work are shown in the upper section of each panel. For comparison, the middle section shows the outcomes of [Farmer et al. \(2023\)](#), determined using the two-parameter explodability criterion of [Ertl et al. \(2016\)](#), while the lower section shows those of [Boccioli & Roberti \(2026\)](#), calculated using neutrino-driven explosion simulations with time-dependent mixing-length theory. Right: Explosion outcomes for the [Woosley & Heger \(2007\)](#) (top) and [Woosley et al. \(2002\)](#) (bottom) progenitor models from [Ebinger et al. \(2019\)](#) shown at the same integer ZAMS masses. Gaps indicate masses for which no model is available.

with [Farmer et al. \(2023\)](#). A similarly mixed picture is found for the Mb series. All of our models up to and including $23 M_{\odot}$ explode, whereas both [Farmer et al. \(2023\)](#) and [Boccioli & Roberti \(2026\)](#) predict a mixture of successful and failed explosions and frequently disagree over individual models. Between 26 and $37 M_{\odot}$, our results are broadly consistent with [Farmer et al. \(2023\)](#), producing a mixture of outcomes. At the highest ZAMS masses, the three approaches again diverge. Overall, the limited agreement across both progenitor series highlights the strong sensitivity of the explodability landscape to the adopted explosion treatment.

Figure 5 also shows the corresponding explosion outcomes for the [Woosley et al. \(2002\)](#) and [Woosley & Heger \(2007\)](#) progenitor sets, WHW02 and WH07. These models provide a useful comparison as they were used in the original PUSH calibration and nucleosynthesis studies ([Ebinger et al. 2019](#); [Curtis et al. 2019](#)), but were evolved using different stellar-evolution assumptions from the [Farmer et al. \(2023\)](#) models explored here. Since the same explosion prescription is applied, differences between Figures 4 and 5 reflect variations in the pre-collapse progenitor structures produced by the underlying stellar-evolution calculations. At low ZAMS masses, the WHW02 and WH07 progenitors show broadly similar behaviour to the Ms and Mb series, with all models up to and including $22 M_{\odot}$ exploding. The first failed explosion in both the WHW02 and WH07 sets occurs at $23 M_{\odot}$, matching the Ms series. The clearest difference between the progenitor grids appears between $25 M_{\odot}$ and $30 M_{\odot}$, where the WHW02 and WH07 sets produce more successful explosions than the Ms and Mb series. This behaviour is likely related to differences in the compactness distributions of the progenitor grids. In the standard PUSH calibration, k_{push} depends on the bounce compactness $\xi_{2.0}$ and reaches its largest values at intermediate compactness, around $\xi_{2.0} \approx 0.3$ (see figures 2 and 8 of [Ebinger et al. 2019](#)). Progenitor grids containing compactness troughs, or a larger number of models with intermediate compactness in this mass range, therefore receive stronger PUSH heating, allowing a greater fraction of models to explode. The role of this calibration is discussed further in Sec-

tion 3.4. Compactness nevertheless provides only a one-parameter characterisation of the core structure: models with similar $\xi_{2.0}$ can have different density profiles, shell structures, and accretion histories, affecting their explosion outcomes and yields. We examine such structural differences in Section 4.2.

3.2 Explosion properties as a function of ZAMS mass

In Figure 6, we compare the remnant masses obtained for the Ms and Mb progenitor series in this work with those of [Farmer et al. \(2023\)](#) and [Boccioli & Roberti \(2026\)](#). Although all three report a remnant mass, this quantity is determined differently in each approach. In both PUSH and the neutrino-driven simulations of [Boccioli & Roberti \(2026\)](#), the mass cut emerges self-consistently from the hydrodynamic evolution. PUSH retains the proto-neutron star within the computational domain. In contrast, [Farmer et al. \(2023\)](#) excise the material interior to the mass coordinate where the entropy per baryon reaches $s = 4$ before initiating the explosion and define the remnant mass as the compact-object mass at shock breakout, neglecting any fallback. The three methods also adopt different equations of state, with PUSH using HS(DD2) ([Hempel & Schaffner-Bielich 2010b](#)), [Boccioli & Roberti \(2026\)](#) using SFHo ([Steiner et al. 2013](#)), and [Farmer et al. \(2023\)](#) using the standard MESA EOS. These differences in numerical treatments may therefore contribute to small systematic offsets between the calculations.

Despite the differences between the three methods, the predicted remnant masses are remarkably similar at low ZAMS masses when the same progenitor successfully explodes. Below ~ 21 – $22 M_{\odot}$, the remnant masses generally agree to within $\sim 0.15 M_{\odot}$, with [Boccioli & Roberti \(2026\)](#) typically predicting the largest values, [Farmer et al. \(2023\)](#) intermediate values, and PUSH the smallest. At higher ZAMS masses, the predictions diverge more strongly. Where direct comparisons between successful Ms models are possible, PUSH consistently produces less massive remnants than [Farmer et al. \(2023\)](#) and [Boc-](#)

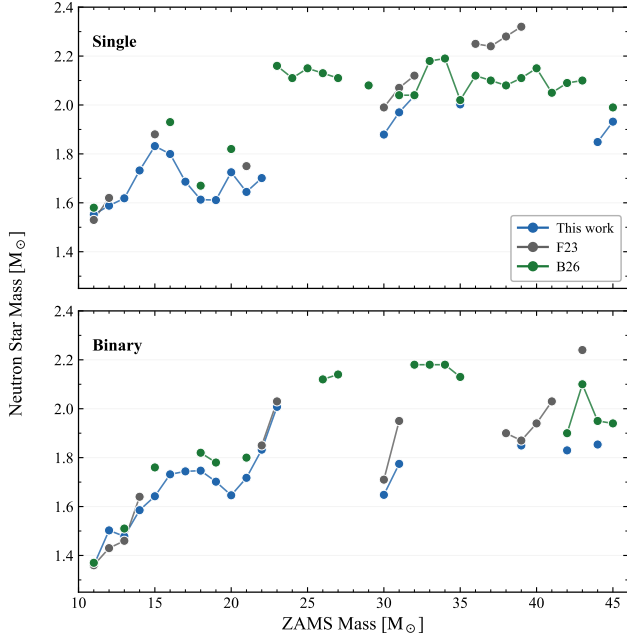


Figure 6. Remnant mass as a function of ZAMS mass for the successfully exploding single-star and binary-stripped models from this work (top), [Farmer et al. \(2023\)](#) (F23; middle), and [Boccioli & Roberti \(2026\)](#) (B26; bottom). Lines connect neighbouring successful models and are broken across gaps corresponding to failed explosions.

[Boccioli & Roberti \(2026\)](#). The same trend is found for the Mb series, although the differences become particularly large above $\sim 40 M_{\odot}$.

Figure 7 shows the explosion energy, remnant mass, and explosion time of the successfully exploding Ms and Mb models as a function of ZAMS mass, with the compactness of each model indicated by the colour bar. The successfully exploding WHW02 and WH07 models from [Ebinger et al. \(2019\)](#) are included for comparison. Overall, the Ms and Mb models occupy similar regions of parameter space, with explosion energies of ~ 0.65 – $1.3 B$, remnant masses of ~ 1.5 – $2.0 M_{\odot}$, and explosion times of ~ 0.35 – 0.65 s. At low ZAMS masses, the Ms and Mb series produce similar remnant masses and explosion times. The main difference occurs between 11 and $14 M_{\odot}$, where the Mb models produce slightly lower explosion energies. These models also have lower compactness values, corresponding to lower values of k_{push} and therefore weaker additional heating. Between 15 and $22 M_{\odot}$, the two series give very similar values for all three properties. Larger differences emerge among the more massive exploding models, particularly above $\sim 30 M_{\odot}$, where the successful Ms models generally produce lower explosion energies, larger remnant masses, and later explosion times than the corresponding Mb models. The Ms models in this range also typically have lower compactness values. We note, however, that the pre-supernova evolution of massive stars is more uncertain in this mass range (e.g. [Hirschi 2015](#)), and the outcome of core collapse can be sensitive to the details of the adopted stellar-evolution prescriptions (e.g. [Sukhbold et al. 2016b](#)).

The explosion properties of the Ms and Mb models are broadly consistent with those obtained previously for the WHW02 and WH07 progenitor sets, although the latter generally explode earlier. This likely reflects differences in both the underlying pre-collapse structures and in the exact evolutionary stage at which the progenitors are mapped into the hydrodynamic simulations. Finally at a fixed ZAMS mass, the differences between the single-star and binary-

stripped progenitors can be comparable to the scatter among the different progenitor sets. We therefore examine the same quantities as a function of compactness in the following section.

3.3 Explosion properties as a function of compactness

Figure 8 shows the explosion energy, remnant mass, and explosion time as a function of the compactness at bounce, $\xi_{2,0}$. Compared with ZAMS mass, compactness reveals clearer trends across all three explosion properties. The strongest is found for the remnant mass: models with larger $\xi_{2,0}$ generally produce more massive neutron stars. This is expected as more compact progenitors sustain higher mass-accretion rates onto the PNS prior to shock revival, allowing the PNS to accrete more material before a successful explosion is launched ([Perego et al. 2015](#)).

The explosion energy also shows a broad dependence on compactness. In both series, the most energetic explosions occur at intermediate compactness, $\xi_{2,0} \sim 0.25$ – 0.4 , while models at both lower and higher compactness generally produce less energetic explosions. This behaviour is closely related to the standard PUSH calibration, in which the compactness-dependent k_{push} reaches its largest values at intermediate $\xi_{2,0}$ and decreases towards both lower and higher compactness. The resulting energy trend therefore partly reflects the imposed compactness-dependent heating prescription, although substantial scatter remains at fixed compactness.

The explosion time shows a weaker dependence on compactness. Models with larger $\xi_{2,0}$ generally explode later than the lowest-compactness models, although there is considerable scatter within this relation. The scatter in remnant mass also varies with compactness: it is largest at low and intermediate compactness, while above $\xi_{2,0} \sim 0.4$ the remnant masses becomes more tightly clustered across the model sets. This differs from the behaviour reported by [Boccioli & Roberti \(2026\)](#), who find substantially larger scatter in the mass cut of the high-compactness Ms progenitors from [Farmer et al. \(2023\)](#).

We note that in general the Ms and Mb models also show substantial overlap in compactness trends with the WHW02 and WH07 models, indicating that binary-stripped progenitors do not occupy a separate region of PUSH explosion parameter space. Compactness therefore provides a useful organising parameter for the explosion properties, but it does not capture all aspects of the pre-collapse structure. Models with similar $\xi_{2,0}$ can still differ in their density profiles, shell locations, and accretion histories, producing differences in both the timing and energetics of shock revival. In particular, [Boccioli & Roberti \(2026\)](#) find that the explodability of lower-compactness progenitors depends strongly on the density drop associated with the Si/O interface, whereas high-compactness explosions are less often triggered by accretion of this interface and instead arise from increasingly efficient neutrino heating. We also find that differences in the Si- and O-burning regions have an important impact on the resulting nucleosynthetic yields, which we discuss further in Section 4.

3.4 Sensitivity to the compactness-dependent k_{push} calibration

The comparisons above show that Ms and Mb progenitors with the same ZAMS mass can have different explosion outcomes. These differences cannot be interpreted from ZAMS mass alone as in the standard PUSH calibration, the strength of the additional heating is itself progenitor dependent. The parameter k_{push} is set by the compactness at bounce, so differences in compactness between an Ms and Mb progenitor affects the amount of additional PUSH heating they receive and therefore their hydrodynamic evolution.

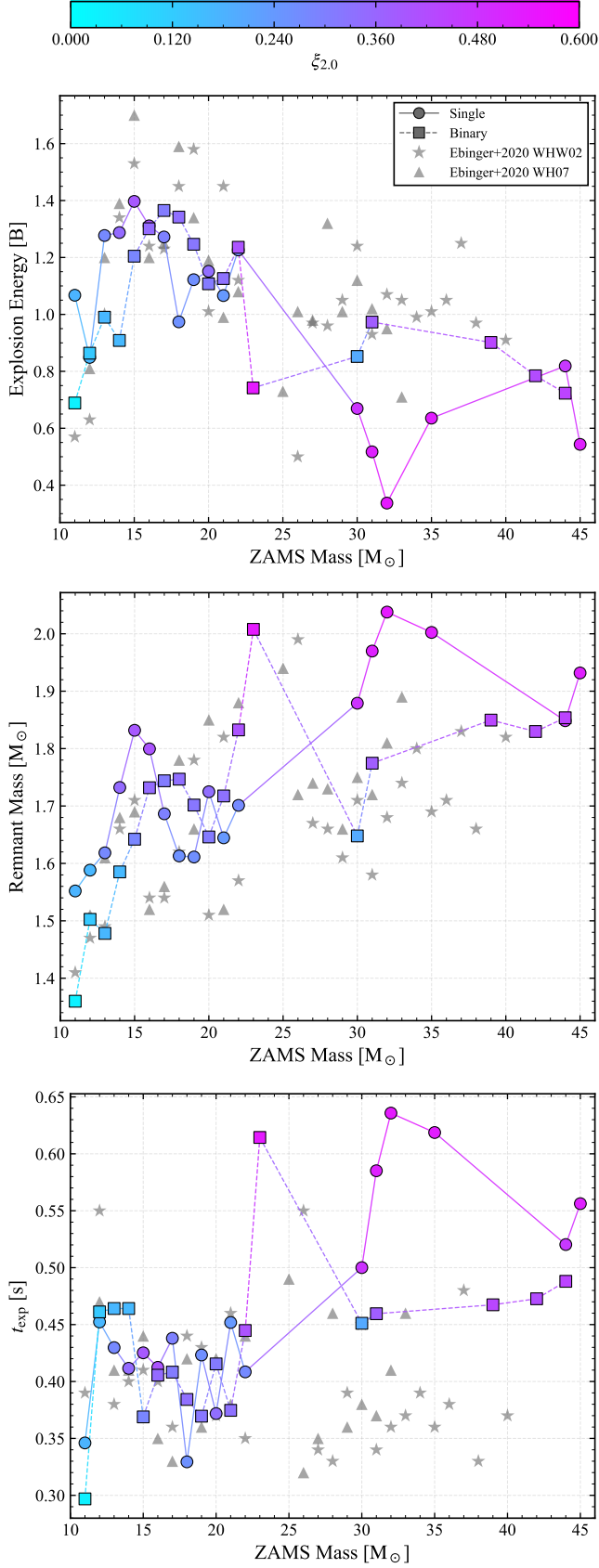


Figure 7. Explosion energy (top), neutron-star remnant mass (middle), and explosion time (bottom) as functions of ZAMS mass for the exploding Ms and Mb models. The WHW02 and WH07 models from Ebinger et al. (2019) are included for comparison. Symbol colour denotes $\xi_{2.0}$ at bounce.

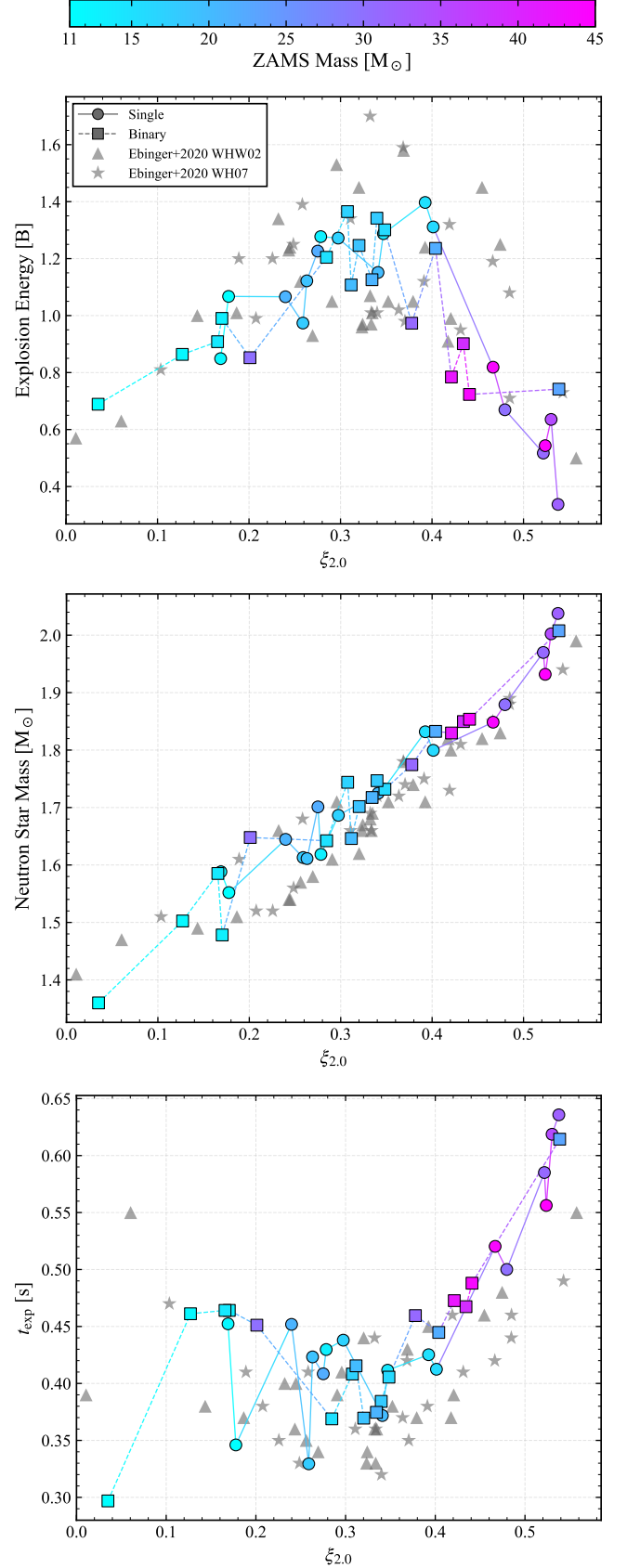


Figure 8. Explosion energy (top), neutron-star remnant mass (middle), and explosion time (bottom) as functions of $\xi_{2.0}$ at bounce for the exploding Ms and Mb models. The WHW02 and WH07 models from Ebinger et al. (2019) are included for comparison. Symbol colour denotes the ZAMS mass.

To test how strongly the explosion outcome depends on this calibration, we considered the first pair with different outcomes, Ms23 and Mb23. Their bounce compactnesses are $\xi_{2,0} = 0.64$ and 0.54 , respectively. With the standard calibration, Ms23 fails to explode while Mb23 explodes successfully. We then repeated both calculations while keeping the progenitor structures unchanged but swapping the compactness values used to determine k_{push} . In other words, Ms23 was assigned the PUSH factor corresponding to Mb23, while Mb23 was assigned that of Ms23. The explosion outcomes reversed: Ms23 exploded with the Mb23 PUSH factor, whereas Mb23 failed with the Ms23 value. We performed the same test for the Ms32 and Mb32 pair, for which the standard calculations give the opposite outcome: Ms32 explodes while Mb32 fails. Swapping their PUSH factors again reversed the explosion outcomes.

These tests highlight an important caveat of the adopted parameterisation of k_{push} : for both model pairs considered, changing only the PUSH heating factor is sufficient to reverse the explosion outcome. The predicted explodability therefore depends strongly on the assumption that compactness is the appropriate and dominant quantity with which to calibrate the additional heating. The differences between the Ms and Mb series must therefore be interpreted within the context of this calibration. Comparisons between progenitors with similar compactnesses are consequently particularly valuable, since they reduce differences arising from the parameterisation and instead provide a clearer test of how the differences in progenitor structure affect the explosion properties and nucleosynthetic yields.

4 NUCLEOSYNTHESIS CALCULATIONS

We compute detailed nucleosynthetic yields for all successfully exploding models in the Ms and Mb series. We first briefly describe the dominant nucleosynthetic production channels from light to heavy elements in Section 4.1, before examining the Ms18.00 and Mb18.00 models in detail in Section 4.2. These models are selected as their ZAMS mass is close to that inferred for the progenitor of SN 1987A. Their abridged isotopic yields are presented in Table B1 of Appendix B. We then examine the impact of binary stripping on elemental yields, including comparisons with previous studies and IMF-weighted yields, in Section 4.3. Finally, we investigate how selected isotopic and iron-group yields vary with progenitor compactness and pre-explosion structure in Section 4.4. Full isotopic yield tables, including radionuclide yields, are provided electronically for all successfully exploding models.

4.1 Overview of synthesis channels

The yields of light elements (H–N) are determined primarily by the pre-supernova evolution of the progenitor, through hydrostatic burning, convective mixing, and mass loss. Li, Be, and B are special cases, with significant contributions from other production sites and processes, including classical novae, low-mass stars, and cosmic-ray spallation. Neutrino interactions can also alter the abundances of rare isotopes in the outer ejecta through the ν -process (Woosley et al. 1990; Heger et al. 2005), although these reactions are not included in our post-processing network. As explosive burning has little effect on the bulk abundances of these lighter elements, their final ejecta reflect the composition and amount of material before core collapse. This is particularly important for the comparison between the Ms and Mb models, as binary stripping substantially alters the mass and composition of the outer stellar layers that are ultimately ejected.

The final yields of intermediate-mass elements (IMEs; O–Ca) reflect contributions from both hydrostatic and explosive burning. During the explosion, additional intermediate-mass elements are synthesised primarily through explosive oxygen burning, although only a fraction of the progenitor’s ^{16}O is processed in this way. Much of the O–Ne shell material does not reach temperatures high enough for substantial explosive burning. Consequently, elements such as O, Ne, and Mg retain substantial hydrostatic contributions, whereas the heavier IMEs are more sensitive to the explosion conditions. Abundance ratios such as $[\alpha/\text{Fe}]$ provide a useful way to place these yields in the context of CCSN populations and galactic chemical evolution, where α denotes elements such as O, Ne, Mg, Si, S, Ar, Ca, and Ti.

The iron-group yields (Sc–Zn) originate predominantly from complete and incomplete silicon burning in the deepest ejecta, making them sensitive to the peak temperatures and densities reached, as well as to the location of the mass cut. The electron fraction, Y_e , is especially important in these regions, as neutrino interactions can drive the ejecta away from the pre-supernova composition and strongly influence the relative production of neutron-rich and proton-rich isotopes. More neutron-rich conditions can favour isotopes such as ^{57}Ni relative to ^{56}Ni , while proton-rich material near the mass cut can enable limited νp -process nucleosynthesis and the production of nuclei beyond the iron group. The Fe-group and trans-iron yields therefore provide sensitive diagnostics of the thermodynamic and weak-interaction histories of the innermost ejecta, although their integrated yields may also contain contributions inherited from the pre-supernova composition.

Our nucleosynthesis post-processing adopts a mass resolution of $10^{-3} M_{\odot}$. We exclude the three innermost tracer particles immediately adjacent to the mass cut, as the late-time neutrino-driven wind is poorly resolved, and do not draw conclusions about its detailed composition. In our calculations, material close to the mass cut reaches electron fractions as low as $Y_e \approx 0.35$. Despite these neutron-rich conditions, the ejecta do not achieve the conditions required for a robust r -process (e.g., Farouqi et al. 2010). We also note that our calculations do not include neutrino-electron scattering, which contributes to neutrino thermalisation and spectral down-scattering and may consequently affect the electron fraction of the wind; see Fischer et al. (2012). For a more comprehensive discussion of the relevant caveats and nucleosynthetic production channels in PUSH calculations, see Section 3 of Curtis et al. (2019).

4.2 Two representative models: Ms18.00 and Mb18.00

The top panels of Figure 9 show the pre-explosion compositions of the inner layers of Ms18 and Mb18, with the mass fractions of several α -chain nuclei plotted against enclosed mass. Both progenitors have broadly similar stratification, with Si- and S-rich material immediately outside the iron core, followed by O- and Mg-rich layers. However, the locations and extent of these regions differ, with Mb18 showing sharper transitions between layers. The iron-rich region extends to $\sim 1.6 M_{\odot}$ in Ms18 and $\sim 1.7 M_{\odot}$ in Mb18, beyond which the iron-group mass fraction falls below 10^{-3} . The Si-, S-, and Ar-rich layers extend farther outwards in Ms18, becoming subdominant to O- and Mg-rich material at $\sim 2.2 M_{\odot}$, compared with $\sim 1.9 M_{\odot}$ in Mb18. The differences in pre-explosion structure become more pronounced farther out. In Mb18, the O-rich region declines sharply at $\sim 4 M_{\odot}$ and is followed by a relatively small He-rich layer, with no H-rich envelope. Ms18 instead retains a more extended O-rich region followed by substantial He- and H-rich outer layers. These outer layers are not shown in full in Figure 9, which focuses on the inner regions. Binary stripping therefore modifies not only the total mass

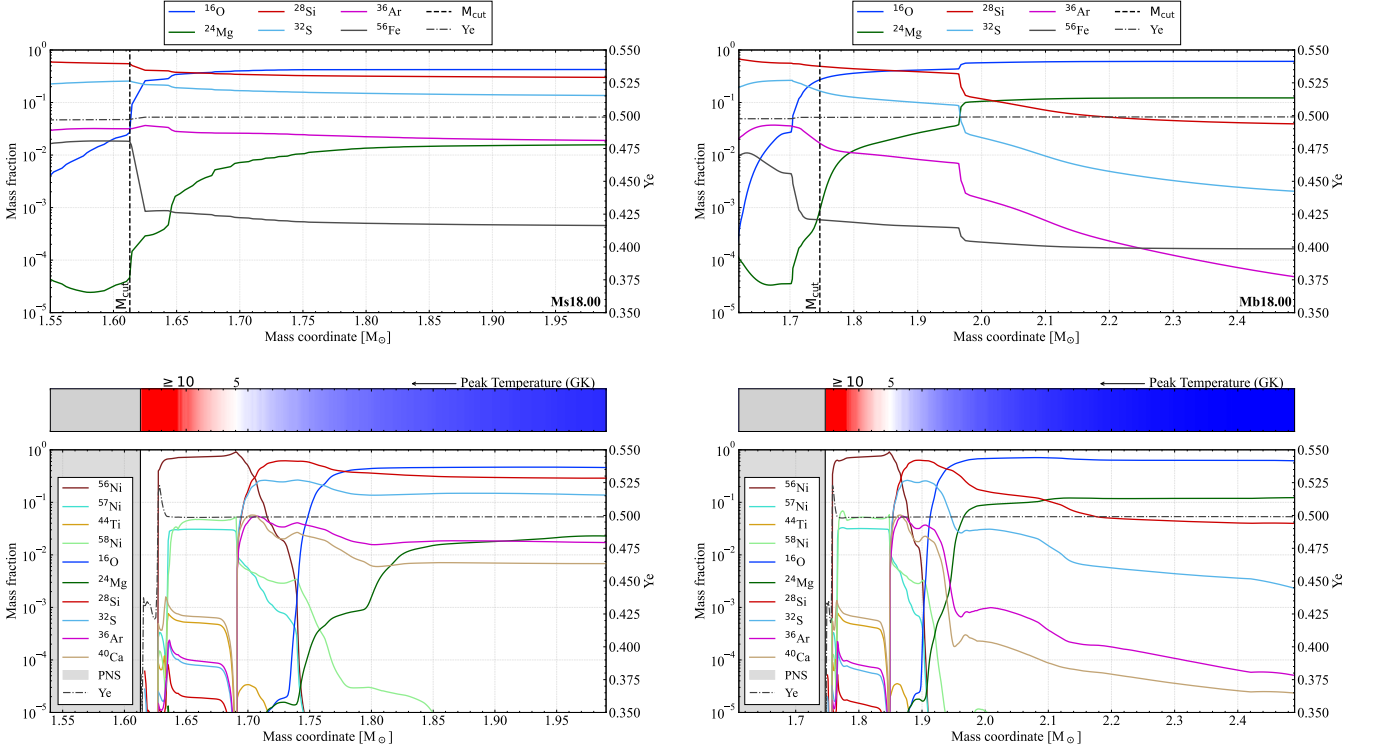


Figure 9. Top panels: pre-explosion composition of the inner layers of the Ms18.00 (left) and Mb18.00 (right) models. Bottom panels: final composition following explosive nucleosynthesis, with the colour scale indicating the peak temperature attained by each layer. The vertical dashed lines mark the mass cuts, while the dot-dashed curves show the corresponding electron-fraction profiles.

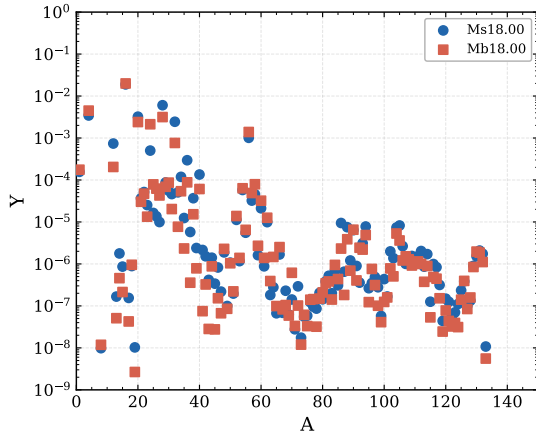


Figure 10. Mass-integrated final abundances Y as a function of mass number A for models Ms18.00 (blue circles) and Mb18.00 (red squares).

and compactness of the progenitors, but also the composition and extent of the layers available for explosive processing and ejection.

We note that the PUSH simulations follow the collapse, core bounce, shock revival, and early shock propagation through the inner progenitor. The extended envelopes are therefore excluded from the hydrodynamic calculations and reattached afterwards (see [Curtis et al. 2021](#)). The resulting ejecta can then be evolved with radiation-hydrodynamics and Monte Carlo radiative-transfer codes, such as SNEC ([Morozova et al. 2015](#)) and SUPERNU ([Wollaeger et al. 2013](#)), to generate synthetic observables.

The bottom panels of Figure 9 show the post-processed composi-

tions of the innermost ejecta exterior to the mass cut, together with the peak temperature and electron fraction reached by each layer. In both models, the innermost ejecta are dominated by iron-group isotopes, particularly ^{56}Ni , ^{57}Ni , and ^{58}Ni . The stable ^{58}Ni -rich material is concentrated interior to the ^{56}Ni -rich region, although its distribution extends to somewhat larger mass coordinates and higher mass fractions in Ms18 than in Mb18. Such differences could influence the synthetic observables, particularly the line profiles associated with stable and radioactive Ni, although detailed non-local thermodynamic equilibrium (NLTE) radiative-transfer calculations would be required to quantify this effect (e.g., [Dessart 2025](#)).

In both models, we find that the mass cut lies within the Si/O-rich shell, but at a larger enclosed mass in Mb18. This is consistent with previous PUSH investigations (e.g., [Ebinger et al. 2019](#); [Curtis et al. 2019](#)) and the trend seen in Section 3.3, where more compact progenitors generally produce larger remnant masses. At bounce, the two models have compactnesses of $\xi_{2.0} = 0.26$ for Ms18 and $\xi_{2.0} = 0.34$ for Mb18. The larger mass cut in Mb18 consequently leaves less of the innermost Si-rich material in the ejecta. Selected isotopic yields of both models are provided in Table B1; negative values in the explosive-yield columns indicate isotopes that undergo net destruction during the explosion.

Figure 10 shows the integrated final abundances of layers reaching peak temperatures above 1.75 GK, corresponding to processed masses of $0.72 M_{\odot}$ and $0.68 M_{\odot}$ for Ms18 and Mb18, respectively. Both models show broadly similar abundance patterns, with most isotopic abundances agreeing to within an order of magnitude. The largest differences occur for light and intermediate-mass nuclei, reflecting variations in the initial composition and stratification, and in the material undergoing explosive O and incomplete Si burning ($24 \leq A \leq 44$). Initially, Ms18 and Mb18 contain $0.294 M_{\odot}$ and

0.324 $M_{\odot}$ of ^{28}Si , respectively, with Si-rich material more centrally concentrated in Mb18 (Figure 9). The larger mass cut in Mb18 therefore removes a greater fraction of this Si-rich material from the ejecta, while the material that is ejected also undergoes more extensive explosive processing. This results in the net destruction of 0.104 $M_{\odot}$ and 0.178 $M_{\odot}$ of ^{28}Si , and final yields of 0.191 $M_{\odot}$ and 0.146 $M_{\odot}$ for Ms18 and Mb18, respectively. Thus, despite its larger initial ^{28}Si mass, Mb18 ejects less ^{28}Si . The higher abundance at $A = 56$ in Mb18 reflects its larger ^{56}Ni yield, consistent with more extensive complete Si burning.

4.3 Impact of Binary Stripping on Elemental Yields

In this section, we investigate the impact of binarity on the nucleosynthetic yields by comparing Ms and Mb progenitors at the same ZAMS mass. We first place our nucleosynthesis calculations in the context of previous studies. Figure 11 compares the yields obtained in this work with those of Farmer et al. (2023) and Boccioli & Roberti (2026). We focus on the Ms11 and Mb11 models, as 11 $M_{\odot}$ is the only ZAMS mass for which both the model series produce successful explosions in all three treatments. The three calculations employ substantially different explosion and nucleosynthesis treatments. We post-process the thermodynamic trajectories with CFNET using a network of 2902 isotopes extending to ^{211}Eu , including neutrino and antineutrino captures. Boccioli & Roberti (2026) post-process their explosions with SKYNER using 1500 isotopes up to $A = 100$, without neutrino interactions during the post-processing, while Farmer et al. (2023) employs the smaller 162-isotope MESA network up to ^{64}Zn .

Despite differences in approach, all three recover similar yield patterns, with most individual cases agreeing to within a factor of a few. The largest differences occur for the iron-group elements, particularly Zn. As noted by Farmer et al. (2023), Zn lies at the upper boundary of their reaction network, causing material to accumulate in Zn isotopes that would otherwise be processed to heavier nuclei. More generally, iron-group elements are synthesised predominantly in the innermost ejecta through complete and incomplete Si burning, making their yields particularly sensitive to the explosion dynamics, mass cut, thermodynamic conditions, and local electron fraction. Boccioli & Roberti (2026) compared previous PUSH calculations (Curtis et al. 2019) with their GRID+ explosion and nucleosynthesis calculations, and found that differences between the treatments can be traced to contributions from the νp -process, weak r -process elements, and the adopted NSE threshold. Despite these differences, when all three treatments agree that a given model explodes, the predicted yields agree remarkably well, with all three showing that binary stripping can substantially alter the yields of particular elements.

Figure 12 shows the ratio of final elemental yields from binary-stripped progenitors to their single-star counterparts across the ZAMS-mass range. These yields reflect both pre-supernova evolution and subsequent reprocessing during collapse and explosion. Single-star progenitors typically have larger yields of lighter elements, which are predominantly produced during hydrostatic evolution and reside in the outer layers. However, binarity also strongly affects the yields of heavier elements, with clear differences for elements from S to Sc in the lowest ZAMS-mass models (11–14 $M_{\odot}$). This is particularly interesting as S, Ar, and Ca are major products of explosive O burning and incomplete Si burning, while the production of Sc probes deeper and hotter layers, where it is largely produced through the decay of ^{45}Ti synthesised in high Y_e regions. This behaviour is consistent with Farmer et al. (2023), who found particularly strong effects of binarity on yields from Al to Sc, while Boccioli & Roberti (2026) also found sensitivity of iron-group yields to the adopted explosion treatment.

To account for differences in total pre-explosion mass, we normalise each yield by the corresponding progenitor’s pre-explosion yield in Figure 13. The lighter elements up to Al remain broadly similar, except in the 44 $M_{\odot}$ progenitor, where a small increase in N produces a considerably larger relative enhancement. Between Si and Sc, the single-star models typically show greater relative enhancement, whereas at intermediate ZAMS masses the binary-stripped models can show greater enhancement from Cr to Fe. The heaviest elements, which are particularly sensitive to the explosion treatment, generally show only slight enhancement in the binary-stripped models, with Zn in the 13 $M_{\odot}$ progenitor the clearest exception. Zn is produced in complete Si burning and in material with $Y_e \gtrsim 0.5$ close to the mass cut. Mb13 contains approximately twice as much Si-rich material near the mass cut as Ms13, partly accounting for its enhanced Zn production. However, differences in compactness and the associated PUSH heating term complicate these trends (see Section 4.4 for a discussion of their dependence on compactness).

Figure 14 shows the ratio of final elemental abundances relative to Fe in each binary-stripped model to those in its corresponding single-star counterpart. Normalising to Fe partly removes differences in the overall scale of the yields and highlights variations in the relative abundance patterns. We find no systematic enhancement across the full mass range; instead, the patterns vary considerably with ZAMS mass. At the lowest masses (11, 13, and 14 $M_{\odot}$), the binary-stripped models generally show enhanced intermediate-mass and several iron-group element abundances relative to Fe. At higher masses, abundances relative to Fe are more commonly lower in the binary-stripped models, with several intermediate-mass elements enhanced in the single-star models at 44 $M_{\odot}$, although considerable scatter remains. This further demonstrates that binarity does not produce a uniform shift in the abundance pattern, with differences depending on both ZAMS mass and the individual element. We also compare several representative species with their corresponding solar abundance ratios in Section 5.3, providing further context for the predicted yields.

We conclude this subsection by assessing the population-level impact of binarity on nucleosynthetic yields, weighting each progenitor by a Salpeter initial mass function (IMF; Salpeter 1955), normalised over the progenitor grid. The yields include pre-supernova winds and final PUSH ejecta for successful explosions, but only wind yields for failed explosions. Following Boccioli & Roberti (2026), we exclude the four binary-stripped progenitors that did not reach core collapse (24, 25, 28, and 29 $M_{\odot}$). We assume the same initial mass distribution for both populations. Figure 15 shows that the IMF-weighted yields of the two populations remain broadly similar, although binarity produces the largest differences for elements from Cl to Ti, with K enhanced by a factor of ~ 3.4 in the binary-stripped population. Farmer et al. (2023) linked enhanced pre-supernova production of ^{39}K in low-mass binary-stripped progenitors to the extent of their Si-rich layers. The greater weight assigned to these progenitors by the Salpeter IMF amplifies their contribution to the integrated K yield. Recently, Rodríguez et al. (2023) inferred a mean CCSN Fe yield of $0.058 \pm 0.007 M_{\odot}$, subsequently used by Weinberg et al. (2024) to anchor the absolute scale of CCSN nucleosynthetic yields. Our IMF-weighted Fe yields are 0.0747 $M_{\odot}$ and 0.0589 $M_{\odot}$ for the single-star and binary-stripped populations, respectively. The binary-stripped population therefore shows closer agreement with this benchmark.

4.4 Trends in nucleosynthetic yields with compactness

We now examine how the yields of key isotopes and iron-group elements in the exploding models depend on progenitor compactness, while also considering differences associated with the pre-explosion

Table 1. Progenitor, explosion, and nucleosynthetic properties of the Ms18 and Mb18 models. The table columns give the ZAMS mass, $\xi_{2.0}$ at bounce, total progenitor mass at collapse, iron-core mass, carbon–oxygen core mass, helium-core mass, hydrogen-rich envelope mass, explosion energy, remnant mass, and the nucleosynthetic yields of selected isotopes. The iron-core mass is defined by the outer boundary of the central region with $Y_e < 0.495$. The CO-core mass is the enclosed mass at the outer boundary where the helium mass fraction satisfies $X_{\text{He}} \leq 0.2$. The helium-core mass is the enclosed mass at the outer boundary where the hydrogen mass fraction satisfies $X_{\text{H}} \leq 0.2$. The hydrogen-rich envelope mass is the mass beyond the helium core.

Model	M_{ZAMS} ($M_{\odot}$)	$\xi_{2.0}$ —	M_{prog} ($M_{\odot}$)	M_{Fe} ($M_{\odot}$)	M_{CO} ($M_{\odot}$)	M_{He} ($M_{\odot}$)	M_{env} ($M_{\odot}$)	E_{expl} (B)	M_{remn} ($M_{\odot}$)	^{56}Ni ($M_{\odot}$)	^{57}Ni ($M_{\odot}$)	^{58}Ni ($M_{\odot}$)	^{44}Ti ($M_{\odot}$)
Ms18	18	0.26	10.08	1.49	5.50	7.41	2.67	0.97	1.61	5.63E-02	1.82E-03	2.59E-03	2.86E-05
Mb18	18	0.34	5.74	1.52	4.31	5.74	0.00	1.34	1.75	7.78E-02	2.78E-03	4.48E-03	3.75E-05

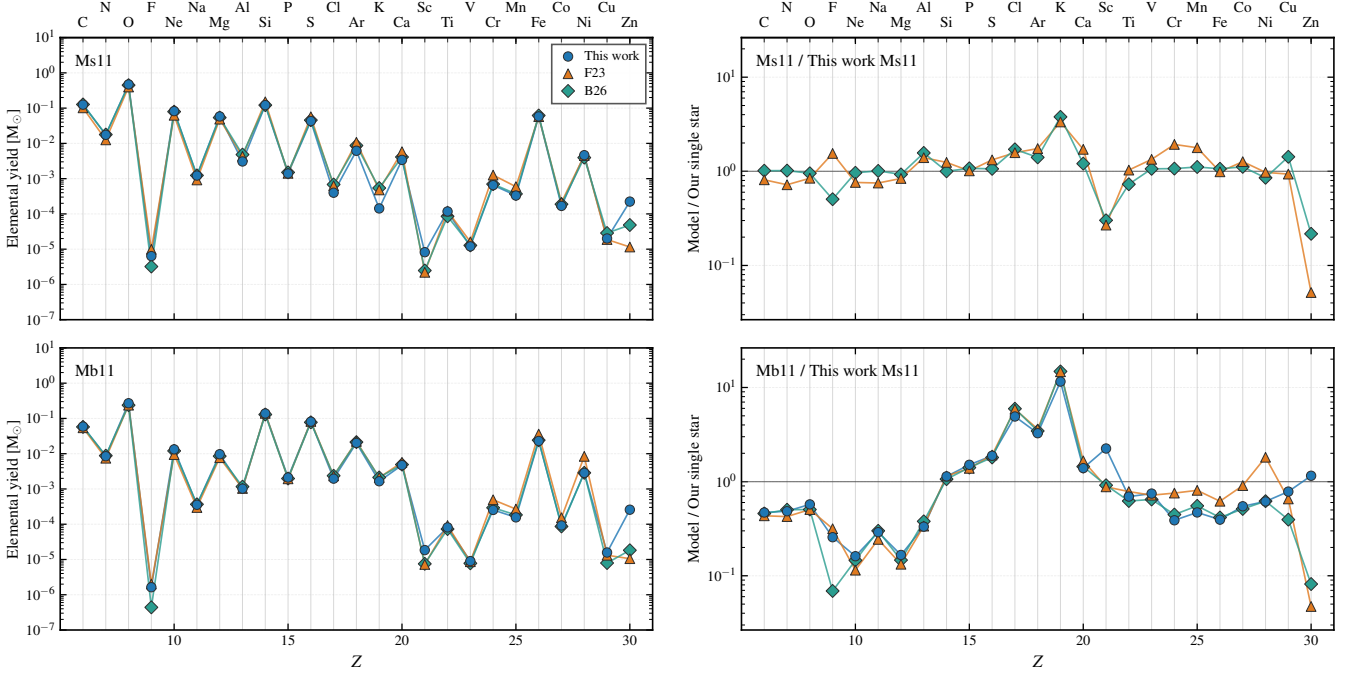


Figure 11. Elemental yields for the Ms11 (top) and Mb11 (bottom) models compared with those of Farmer et al. (2023) and Boccioli & Roberti (2026). The left-hand panels show the absolute elemental yields, whereas the right-hand panels show the corresponding yields relative to the Ms11 yields from this work (our single-star model). Note that in the upper-right panel only the Farmer et al. (2023) and Boccioli & Roberti (2026) models are shown and that the dashed horizontal line marks equal yields with the Ms11 model from this work.

structure. Figure 16 shows the yields of ^{56}Ni , ^{57}Ni , ^{58}Ni , and ^{44}Ti as a function of $\xi_{2.0}$, with the models colour-coded by ZAMS mass. This allows the yield trends to be considered alongside the progenitor structures shown in Figures 2 and 3. The yields of ^{56}Ni and ^{44}Ti generally increase with compactness, while ^{57}Ni follows a similar trend but with greater scatter. At higher compactness, the single-star and binary-stripped models investigated here show somewhat more scatter in these three isotopes than the previous PUSH calculations (Curtis et al. 2019). The ^{58}Ni yields exhibit substantial scatter across the entire compactness range. These variations highlight the sensitivity of the yields to differences in progenitor structure that are not captured by compactness alone.

The scatter at fixed compactness is particularly apparent among the single-star models. For example, Ms35 and Ms45 both have $\xi_{2.0} \simeq 0.52$, yet Ms45 produces almost twice as much ^{56}Ni . This difference reflects their pre-explosion structures: Ms45 has a larger and more compressed Si-rich region available for explosive processing, allowing more material to undergo Si burning and contribute to the radioactive Ni yield. The binary-stripped sequence also shows differences in yields at comparable compactness, although these are generally smaller than those among the single-star models.

The weakest dependence on compactness is found for the stable, neutron-rich isotope ^{58}Ni . Although the most compact models tend, on average, to produce somewhat larger ^{58}Ni yields, the models do not follow a single well-defined trend. This is broadly consistent with Curtis et al. (2019), who found that neutron-rich isotope production depends on the local electron fraction, Y_e . In their calculations, ^{58}Ni yields separate into multiple branches associated with different Y_e conditions, with lower Y_e generally favouring greater production of neutron-rich isotopes such as ^{57}Ni and ^{58}Ni .

Figure 17 shows the final elemental yields of Ti, Cr, Mn, and Ni as a function of compactness, with the models colour-coded by ZAMS mass. These trends can be understood through the dominant isotopes and burning regimes responsible for their production. Ti and Cr are produced predominantly through ^{48}Cr and ^{52}Fe , nuclei with equal numbers of protons and neutrons, which decay to ^{48}Ti and ^{52}Cr , respectively. Most Mn is produced as ^{55}Co during incomplete Si burning, while Ni is produced both directly as stable Ni isotopes and through the decay of nuclei such as ^{60}Zn and ^{60}Cu during complete Si burning (see Figure 4 of Curtis et al. 2019). The differences in yield trends are particularly pronounced among the highest-ZAMS-mass models, where they reflect differences in pre-explosion structure. In

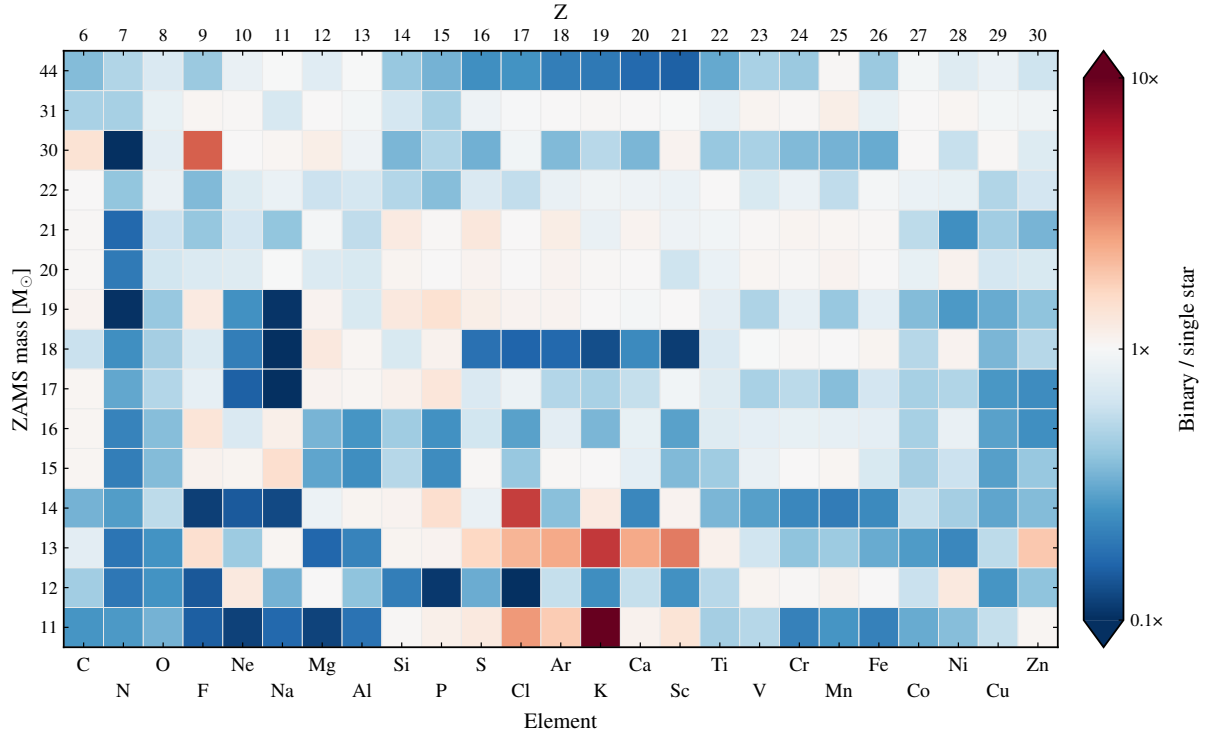


Figure 12. Ratio of the total elemental yields from the binary-stripped (Mb) progenitors to their corresponding single-star (Ms) progenitors as a function of ZAMS mass and element. Red indicates enhanced production in the binary-stripped model, while blue indicates enhanced production in the single-star model. Ratios are shown only for models for which both the Ms and Mb progenitors produce successful explosions.

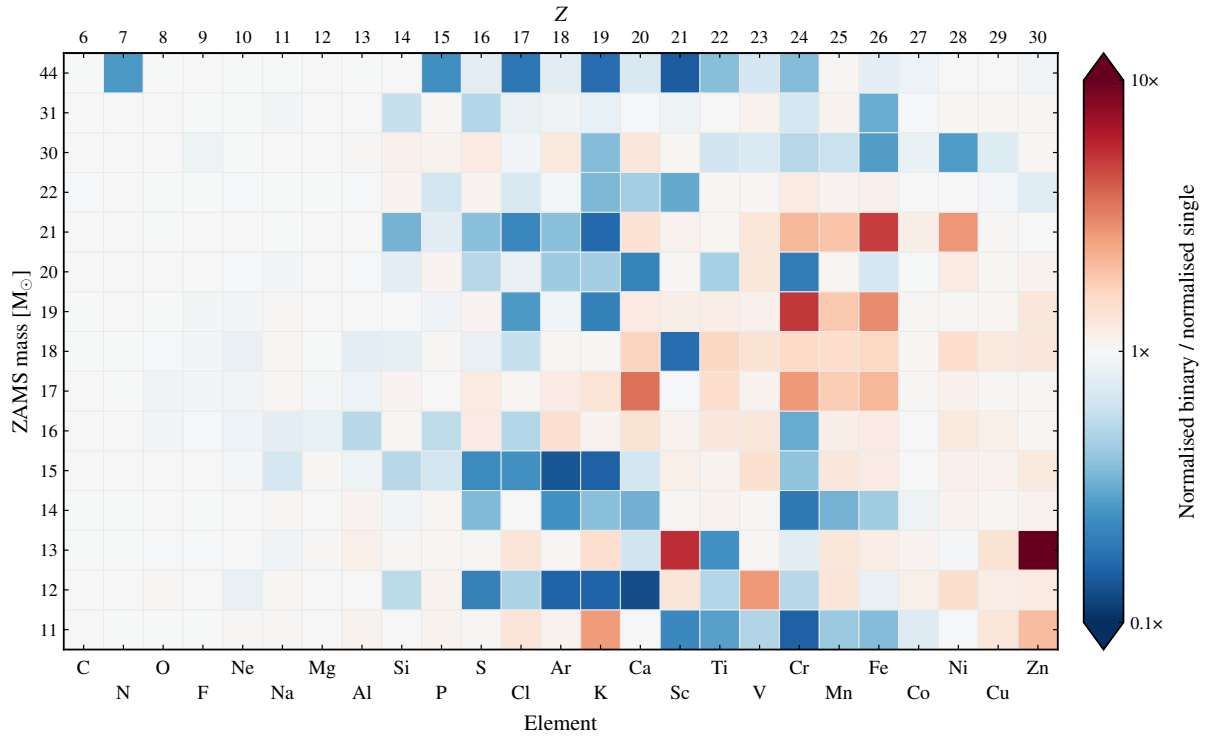


Figure 13. Similar to Figure 12, but with each elemental yield normalised by the corresponding initial progenitor abundance before taking the ratio between the binary-stripped and single-star models.

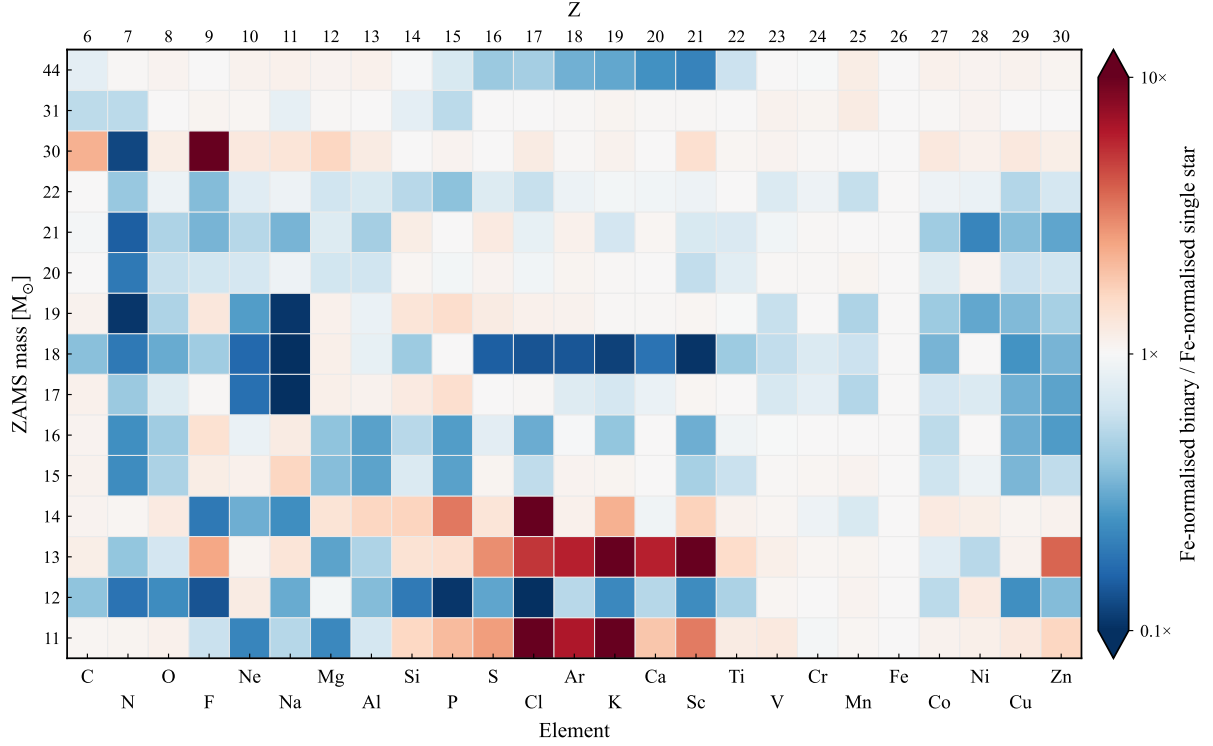


Figure 14. Ratio of the final elemental abundances relative to Fe in the binary-stripped (Mb) progenitors to those in their corresponding single-star (Ms) progenitors as a function of ZAMS mass and element. Red indicates a larger abundance relative to Fe in the binary-stripped model, while blue indicates a larger abundance relative to Fe in the single-star model. Ratios are shown only for models for which both the Ms and Mb progenitors produce successful explosions.

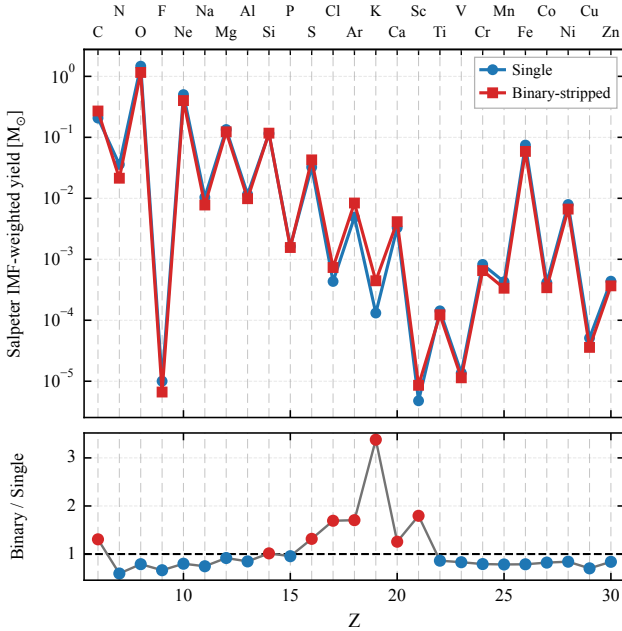


Figure 15. Salpeter IMF-weighted elemental yields for the single-star (Ms) and binary-stripped (Mb) progenitor populations, with the corresponding yield ratio shown in the lower panel. Red points in the lower panel indicate greater production by the binary-stripped population, whereas blue points indicate greater production by the single-star population. The dashed horizontal line marks equal production between the two populations.

Mb44, the Si-dominated region above the mass cut extends from $\sim 1.8 M_{\odot}$ to $\sim 2.3 M_{\odot}$, beyond which O-rich material becomes dominant, whereas in Ms44 the Si-rich region extends to $\sim 2.6 M_{\odot}$. Ms44 therefore has a larger reservoir of Si-rich material available for explosive processing, allowing more material to undergo complete burning and enhancing the production of Ti, Cr, and Ni. Mn follows this behaviour less closely, reflecting its stronger association with incomplete Si burning and its sensitivity to the local electron fraction.

5 COMPARISON WITH OBSERVATIONAL CONSTRAINTS

To place the predicted nucleosynthetic yields in an observational context, we compare the single-star and binary-stripped models with constraints from individual CCSNe in Section 5.1 and abundance patterns in metal-poor stars in Section 5.2. We then examine α -element abundance ratios and their observational constraints in Section 5.3, before comparing our IMF-weighted yields with empirical CCSN [X/Mg] ratios inferred from APOGEE in Section 5.4.

5.1 Comparison with the observed landscape

Radioactive ^{56}Ni is an important energy source in CCSNe, with its decay through ^{56}Co to ^{56}Fe heating the ejecta, influencing their ionisation, spectral evolution, and radioactive tail of the light curve. The ejected ^{56}Ni mass therefore provides an observational constraint on explosive nucleosynthesis. Observationally, this is done by relating the bolometric luminosity during the radioactive tail to the decay energy deposited in the ejecta, although uncertainties in distance, extinction, and date of explosion affect the inferred mass (Bruenn

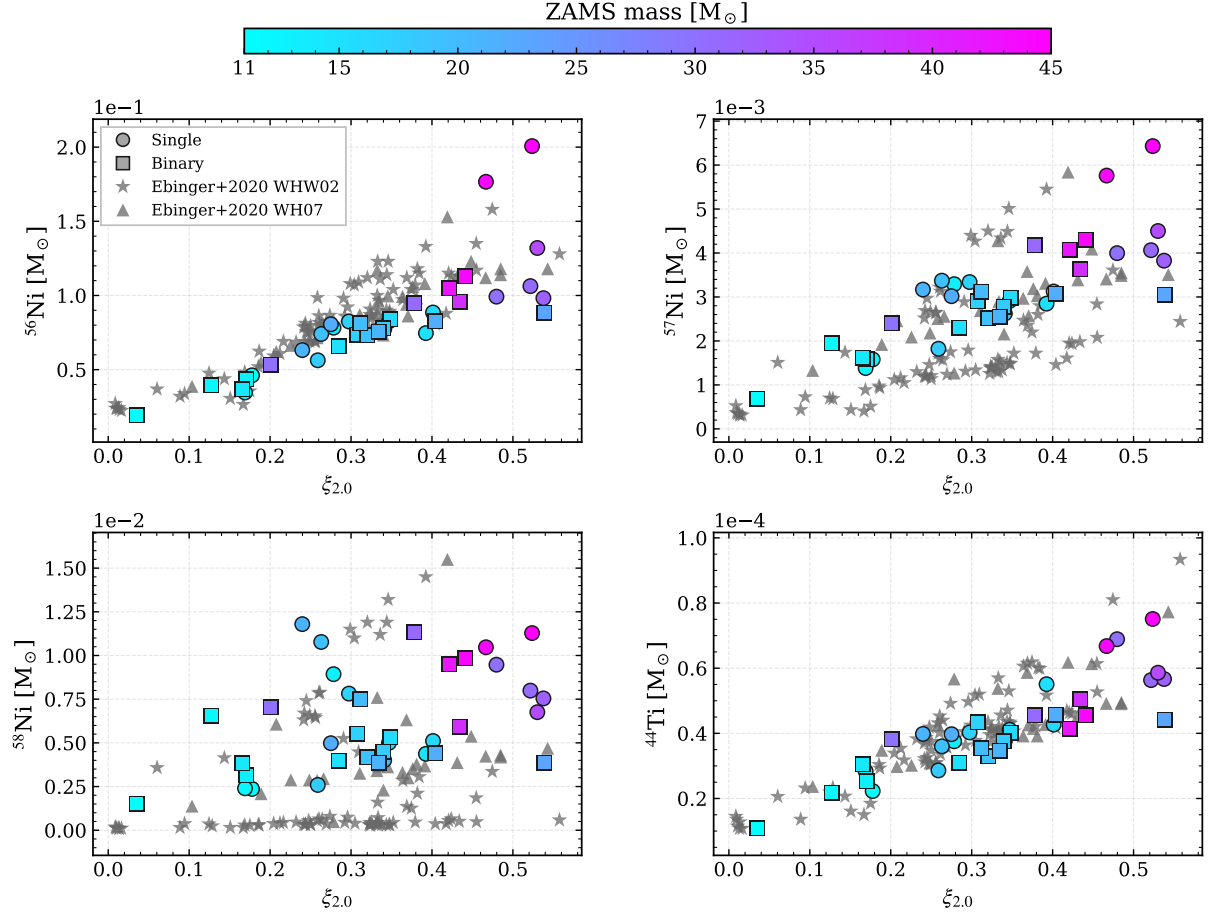


Figure 16. Isotopic yields of ^{56}Ni (top left), ^{57}Ni (top right), ^{58}Ni (bottom left), and ^{44}Ti (bottom right) after explosive processing, shown as a function of compactness for the single-star (blue circles) and binary-stripped (red squares) model series. The corresponding yields from [Curtis et al. \(2019\)](#) are overplotted for the WHW02 and WH07 model series as grey triangles and stars, respectively.

[et al. 2016a](#)). Figure 18 shows the ejected ^{56}Ni mass as a function of ZAMS mass for the single-star and binary-stripped explosion models. For comparison, we include the nucleosynthetic yields calculated by [Curtis et al. \(2019\)](#) for the explosion models investigated in [Ebinger et al. \(2019\)](#). The observational estimates are taken from [Nomoto et al. \(2013\)](#) and [Bruenn et al. \(2016a\)](#), excluding hypernovae. This sample spans inferred progenitor ZAMS masses of $\sim 10\text{--}23 M_{\odot}$ and ^{56}Ni masses of $\sim 0.01\text{--}0.09 M_{\odot}$, with no constraints above $\sim 25 M_{\odot}$.

Both model series broadly cover the observed yield range, although differences emerge at individual ZAMS masses. Between 11 and $15 M_{\odot}$, the binary-stripped models generally produce weaker explosions and smaller ^{56}Ni yields. The Mb11 model, in particular, lies close to the two lowest observationally inferred ^{56}Ni masses, whereas Ms11 ejects considerably more ^{56}Ni . The smaller ^{56}Ni yield of Mb11 is consistent with its lower compactness, in line with the general trends in ^{56}Ni production discussed in Section 4.4. More broadly, our models show less scatter in ^{56}Ni yields than those of [Curtis et al. \(2019\)](#) up to $\sim 23 M_{\odot}$, but substantially greater variation at higher ZAMS masses, where the previous calculations found relatively constant yields. This is particularly evident for Ms44 and Ms45, which produce approximately twice as much ^{56}Ni as similar mass binary-stripped models. As discussed above, this difference is associated with variations in pre-explosion structure and the amount of Si-rich material available for explosive processing.

Figure 19 compares the predicted Ni/Fe ratios and ^{56}Ni yields of the single-star and binary-stripped models with the observational estimates compiled by [Jerkstrand et al. \(2015\)](#) for SN 1987A, SN 2004et, SN 2012aw, and SN 2012ec. The comparison includes models with ZAMS masses between 12 and $21 M_{\odot}$, with marker colour indicating the bounce compactness, $\xi_{2.0}$, and the dashed horizontal line marking the solar Ni/Fe ratio. The low-compactness Mb12 and Mb14 models provide the closest overall agreement with SN 2012ec, reproducing both its supersolar Ni/Fe ratio and its comparatively large ^{56}Ni yield. The Ms12 and Mb13 models produce comparable ^{56}Ni yields, but their predicted Ni/Fe ratios differ from that inferred for SN 2012ec. Several higher-compactness models are consistent with both the ^{56}Ni yields and Ni/Fe ratios inferred for SN 1987A and SN 2012aw. However, no model produces a Ni/Fe ratio low enough to match SN 2004et. Overall, the models span much of the observed range in both quantities.

5.2 Comparison with metal poor stars

Massive stars enriched the early Universe through their winds and subsequent supernovae. Therefore, the atmospheric compositions of long-lived, low-mass, metal-poor stars can preserve the chemical signatures of these early generations of CCSNe, providing observational constraints on predicted nucleosynthetic yields. Numerous

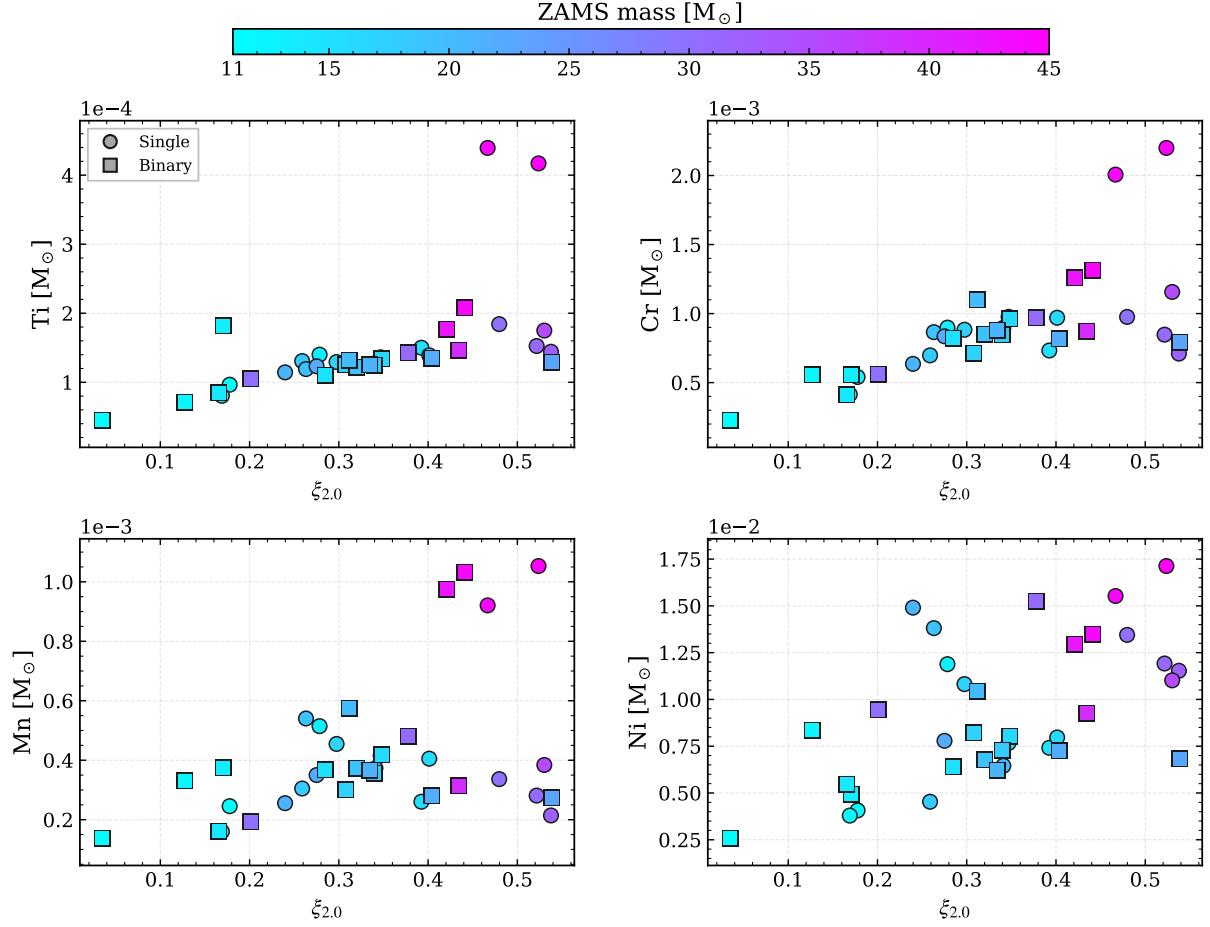


Figure 17. Elemental yields of titanium (top left), chromium (top right), manganese (bottom left), and nickel (bottom right) after explosive processing as a function of compactness for both single-star (blue circles) and binary-stripped (red squares) model series.

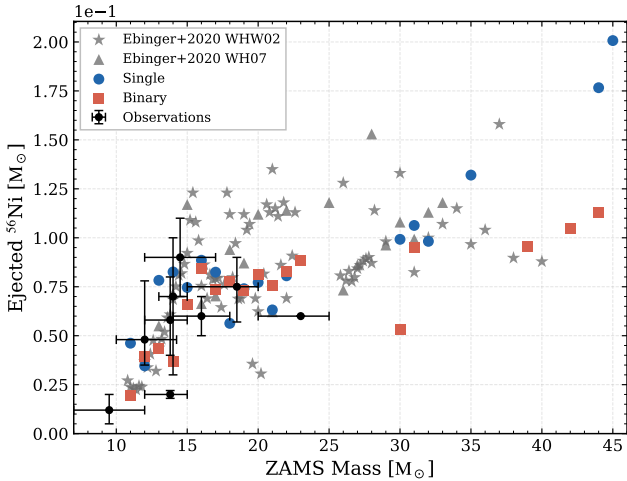


Figure 18. Yields of ^{56}Ni as a function of ZAMS mass for the single-star (blue circles) and binary-stripped (red squares) model series. Observational estimates for faint supernovae are overplotted as black crosses with error bars (see Nomoto et al. 2013; Bruenn et al. 2016b, for further details). The corresponding yields from Curtis et al. (2019) are overplotted for the WHW02 and WH07 model series as grey triangles and stars, respectively for comparison.

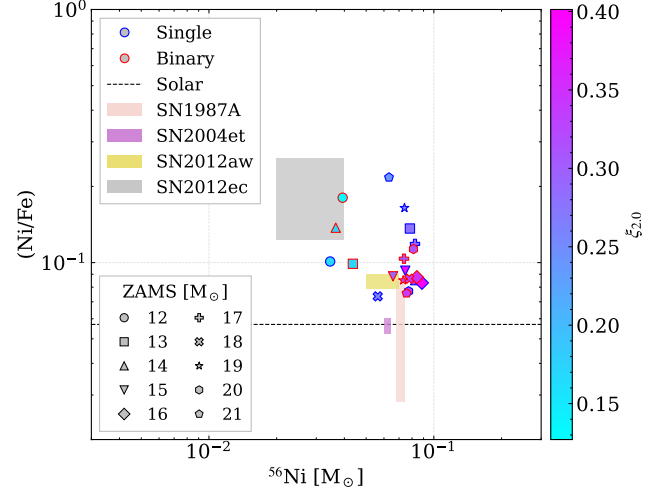


Figure 19. Ni/Fe ratio as a function of ejected ^{56}Ni for SN 1987A, SN 2004et, SN 2012aw, and SN 2012ec (see Jerkstrand et al. 2015). The observations are compared with the single-star and binary-stripped model series, restricted to the ZAMS mass range 12–21 $M_{\odot}$, with marker symbols denoting ZAMS mass. The dashed horizontal line marks the solar Ni/Fe ratio. No uncertainties in the Ni/Fe ratios were reported for SN 2004et and SN 2012aw; these points are therefore shown with error bars comparable to the marker size for visibility.

photometric and spectroscopic surveys have identified metal-poor stars and characterised their atmospheric abundances (e.g. Caffau et al. 2013; Norris et al. 2013; Keller et al. 2014; Li et al. 2015a,b; Howes et al. 2015; Placco et al. 2015; Jacobson et al. 2015; Hansen et al. 2015; Roederer et al. 2016; Abohalima & Frebel 2018; Starkenburg et al. 2017; Andrievsky et al. 2018). More recent studies have further expanded the identification and abundance characterisation of metal-poor stars (e.g. Da Costa et al. 2019; Marino et al. 2019; Venn et al. 2020; Yong et al. 2021; Aoki et al. 2022; Li et al. 2022; Zepeda et al. 2022; Arentsen et al. 2023; Griffith et al. 2023; Martin et al. 2024; Griffith et al. 2026). The very metal-poor main-sequence turnoff star HD 84937 ($[\text{Fe}/\text{H}] = -2.32$) provides a useful benchmark, with detailed abundances derived by Sneden et al. (2016) using improved laboratory transition data for neutral and singly ionised iron-group species. In Figure 20, we compare these abundances with the elemental $[\text{X}/\text{Fe}]$ ratios predicted by our single-star and binary-stripped models. The triangles show the abundances inferred from neutral and singly ionised species, with 3σ uncertainties, while the semi-transparent horizontal segments represent individual progenitor models. We show these separately rather than applying IMF weighting to highlight how agreement with the observed abundances varies within each model series.

Overall, the predicted iron-group abundance patterns broadly agree with that observed in HD 84937, although the level of agreement varies between elements and individual models. The largest spread is seen for Sc and Zn, both of which have traditionally been challenging to reproduce in sufficient quantities in CCSN calculations. Nevertheless, subsets of both the single and binary-stripped models reproduce the observed Sc and Zn abundances. Their production is particularly sensitive to conditions in the innermost ejecta. The largest $[\text{Sc}/\text{Fe}]$ ratios are obtained for the Mb11 and Mb13 models, consistent with the enhanced Sc yields seen in Figure 12. The clearest discrepancy with HD 84937 occurs for $[\text{Cu}/\text{Fe}]$, with our models systematically predicting ratios above the observed value. Cu production is metallicity dependent, and our progenitors have solar metallicity, whereas HD 84937 is very metal-poor, so some disagreement is not unexpected. The low-metallicity PUSH calculations of Ebinger et al. (2020b) obtained Cu abundances consistent with the observations. Finally, both model series produce a range of $[\text{Mn}/\text{Fe}]$ ratios, many of which lie within the observational constraints.

5.3 Comparison of alpha element abundances

We now examine whether single and binary-stripped progenitors produce systematically different alpha-to-iron abundance ratios, focusing on three representative alpha nuclei: ^{16}O , ^{28}Si , and ^{40}Ca . In metal-poor stars the observed surface abundances largely reflect enrichment by the interstellar medium by earlier core-collapse supernovae, particularly when $[\text{Fe}/\text{H}] \lesssim -1$. Metal-poor stars typically exhibit alpha enhancements of $[\alpha/\text{Fe}] \simeq +0.3$ – $+0.5$ dex, although there is considerable scatter between individual elements and stellar populations (Cayrel et al. 2004; Nomoto et al. 2013). At higher metallicities, the delayed contribution of Type Ia supernovae becomes increasingly important and lowers the characteristic alpha-to-iron ratio, as they produce a considerable amount of Fe-group elements. For an isotope ^AX , we calculate its abundance relative to Fe and normalise this ratio to the corresponding solar value. We adopt the solar elemental abundances of Asplund et al. (2009) and the isotopic fractions of Lodders et al. (2009).

Figure 21 shows the resulting abundance ratios. The clearest trend occurs for $[\text{O}/\text{Fe}]$, which generally increases with ZAMS mass, rising from ~ 0.2 – 0.4 dex at $11 M_{\odot}$ to ~ 1.0 – 1.3 dex for the higher-

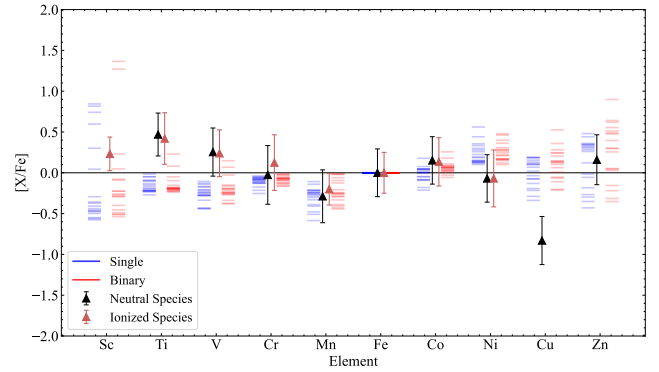


Figure 20. Elemental abundance ratios, $[\text{X}/\text{Fe}]$, for the single-star (blue) and binary-stripped (red) model series, compared to the observed neutral and ionized abundances of HD 84937 (black and red triangles, respectively), with 3σ uncertainties. Model predictions are shown as short horizontal segments, offset to the left (single-star) and right (binary-stripped) of each element for clarity, with each segment corresponding to an individual model. The transparency of the segments is normalized per series such that darker colours indicates a larger number of models predicting a particular abundance ratio, with all models in a series combined producing the full colours shown in the legend.

mass models. Most exploding models between 11 and $22 M_{\odot}$ have ratios of ~ 0.2 – 0.8 dex. This trend reflects the hydrostatic origin of most ejected ^{16}O : as the ZAMS mass and CO-core mass increase, the oxygen-rich layers generally grow. In contrast, the final Fe yield is dominated by the decay of explosively synthesised ^{56}Ni and depends strongly on the explosion properties and mass cut. As shown in Figure 18, the ^{56}Ni yield initially increases with ZAMS mass before broadly levelling off around $15 M_{\odot}$, although substantial variation remains at the highest masses. The continued growth of the oxygen-rich layers without a corresponding increase in Fe therefore explains the general rise in $[\text{O}/\text{Fe}]$. The single-star and binary-stripped models exhibit broadly similar behaviour, typically differing by less than 0.2 dex at the same ZAMS mass. This similarity is consistent with most oxygen residing in the deeper CO-rich layers rather than the outer envelope removed by binary stripping.

The $[\text{Si}/\text{Fe}]$ ratio generally decreases with ZAMS mass, although neither progenitor series follows a smooth trend. Ratios reach ~ 1.0 dex in the lowest mass models, fall to ~ 0.1 – 0.3 dex at 22 – $35 M_{\odot}$, and rise slightly to ~ 0.3 – 0.4 dex in the highest mass explosions. This behaviour reflects the competing contributions to the final ^{28}Si yield. Some is inherited from the progenitor, while additional ^{28}Si is produced through explosive oxygen burning. Material exposed to higher temperatures can instead undergo further processing into Fe-group nuclei. Changes in shock heating and the mass cut therefore affect both the surviving ^{28}Si and the Fe yield, producing a less straightforward dependence on ZAMS mass than for $[\text{O}/\text{Fe}]$.

The $[\text{Ca}/\text{Fe}]$ ratio shows the greatest scatter with ZAMS mass. Below $\sim 15 M_{\odot}$, both progenitor series contain positive and negative ratios, while most intermediate-mass explosions cluster around -0.2 dex, with several of the highest-mass models producing positive values. The production of ^{40}Ca through incomplete silicon burning is sensitive to the amount of ejecta reaching the required temperatures (see figure 4 of Curtis et al. 2019). Stronger shock heating can increase the Ca yield while also processing more material through complete silicon burning into Fe-group nuclei. If the Fe yield increases more rapidly, $[\text{Ca}/\text{Fe}]$ decreases even when the absolute

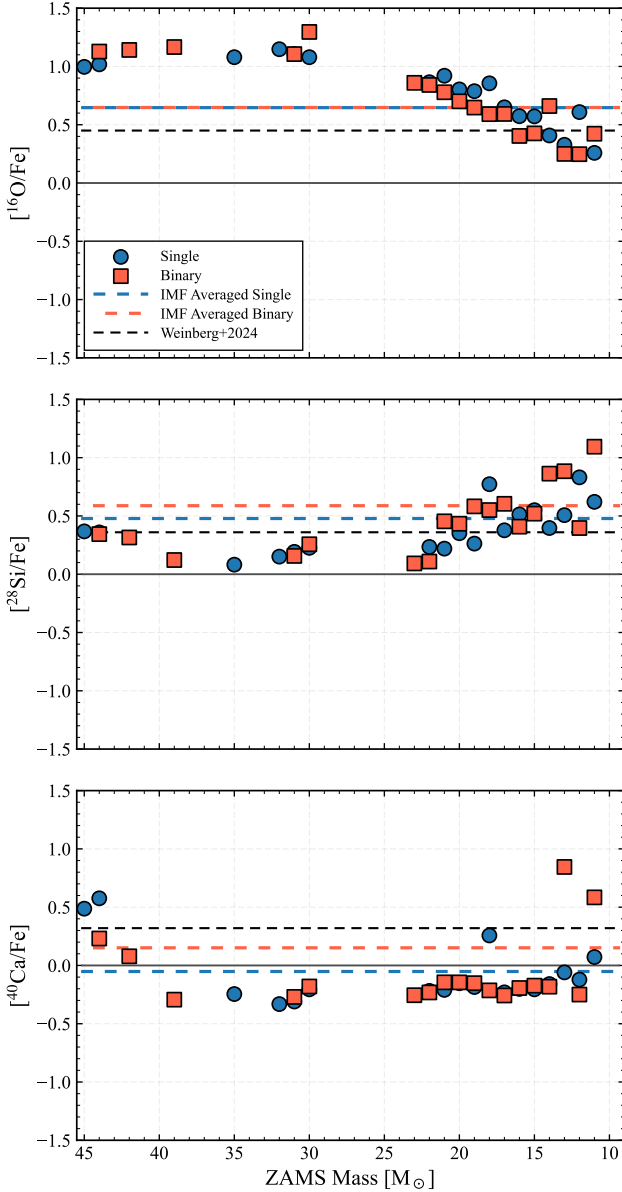


Figure 21. Top to bottom: Abundance ratios of ^{16}O , ^{28}Si and ^{40}Ca to Fe, relative to their solar ratios, for the Ms models (blue circles) and Mb models (red squares). Coloured horizontal lines show the corresponding Salpeter IMF-averaged abundance ratios. Black dashed lines indicate the observational plateau values from Weinberg et al. (2024).

Ca yield rises. The resulting ratios therefore depend on both the thermal history of the ejecta and the final mass cut.

We now consider the IMF-averaged abundance ratios of the two progenitor populations and compare them with the observed low-metallicity $[\alpha/\text{Fe}]$ plateau, which we assume represents pure CCSN enrichment prior to the onset of SN Ia. We adopt O and Si plateaus of 0.45 and 0.36 dex, respectively, as determined in Weinberg et al. (2024) from SDSS DR17 APOGEE data (Majewski et al. 2017), and extend their methods to determine a corresponding plateau value for Ca of 0.32 dex. These values can be compared to the IMF-averaged abundance for our single-star and binary-stripped yield sets, weighting by a Salpeter IMF.

We plot both the IMF averaged abundances and the observed

low-metallicity plateaus as horizontal lines in Figure 21. We find that both the $[\text{O}/\text{Fe}]$ and $[\text{Si}/\text{Fe}]$ are overpredicted in the single and binary models by 0.1 to 0.2 dex, while the $[\text{Ca}/\text{Fe}]$ is underpredicted by 0.37 dex for the single-star models, and 0.17 dex for the binary star models. As noted in Weinberg et al. (2024), the plateau values are uncertain, with estimates of the $[\text{O}/\text{Fe}]$ plateau ranging from 0.3 to 0.6 dex. Within this uncertain range of ± 0.15 dex, our single-star and binary-stripped models are near the upper limit of plausible $[\text{O}/\text{Fe}]$ and $[\text{Si}/\text{Fe}]$ plateaus, while our binary-stripped model is near the lower limit of a plausible $[\text{Ca}/\text{Fe}]$ plateau. The single-star model, however, severely under produces Ca relative to Fe and cannot be reconciled with observations.

5.4 Comparison with empirical $[\text{X}/\text{Mg}]$ abundances

To further compare our single-star and binary-stripped IMF averaged yields with observations, we follow the methodology of Griffith et al. (2021) and measure the offset between the theoretical CCSN abundance ratios and empirical measurements of the CCSN component of observed stellar abundances. The empirical values are taken from Griffith et al. (2024), who fit a two-process model to SDSS DR17 APOGEE abundances (Majewski et al. 2017). The model describe each abundance of each star as the sum of a CCSN and SN Ia process, allowing the CCSN component of the observed abundance space to be separated from the more delayed SN Ia enrichment. Following Griffith et al. (see Appendix B 2021), the two-process model parameters can be translated into an empirical CCSN abundance ratio and compared to the theoretical yields. We define $\Delta[\text{X}/\text{Mg}]_{\text{cc}}$ as the difference between the theoretical and empirical yields, relative to Mg. We use Mg as the reference element as it is produced almost entirely by CCSN and is well measured observationally.

The $\Delta[\text{X}/\text{Mg}]_{\text{cc}}$ for the single-star and binary-stripped yield sets are shown in Figure 22. If $\Delta[\text{X}/\text{Mg}]_{\text{cc}} = 0$, then the model reproduces the empirical APOGEE CCSN yields. Positive/negative values indicate that the models over/underproduce a given element relative to Mg. We find that most elements have a $\Delta[\text{X}/\text{Mg}]_{\text{cc}}$ between -0.25 and 0.15 dex, where O, Na, Al, and Si are overproduced relative to Mg while S, K, Ca, Cr, Mn, Fe, and Co are underproduced in both the single and binary models. For all elements but S, K, and Ca the $\Delta[\text{X}/\text{Mg}]_{\text{cc}}$ for the single and binary models are within 0.1 dex.

The apparent overproduction of α elements O, Al, and Si by ~ 0.15 dex may be a manifestation of the underproduction of Mg in the models, as seen in many CCSN yield sets (e.g., Rybizki et al. 2017). The overproduction of Na, however, is much more drastic, with a $\Delta[\text{X}/\text{Mg}]_{\text{cc}}$ of 0.5 dex for the binary-stripped models and 0.6 dex for single-star models. Griffith et al. (2021) also find a drastic overproduction of Na when comparing yields from Sukhbold et al. (2016a) with observational constraints. Conversely, K is significantly underproduced relative to Mg—but only in the single-star models. S and Ca also show more significant underproduction in the single-star model, but to a lesser degree.

6 DISCUSSION AND CONCLUSIONS

In this work, we have used the PUSH framework to investigate the explodability, remnant properties, and nucleosynthetic yields of the single-star and binary-stripped progenitors of Farmer et al. (2023). We considered 35 single-star and 31 binary-stripped models spanning ZAMS masses of $11\text{--}45 M_{\odot}$, using the standard PUSH calibration (Ebinger et al. 2019) and the CFNET reaction network (Fröhlich et al. 2006) for nucleosynthesis post-processing. We compared our results

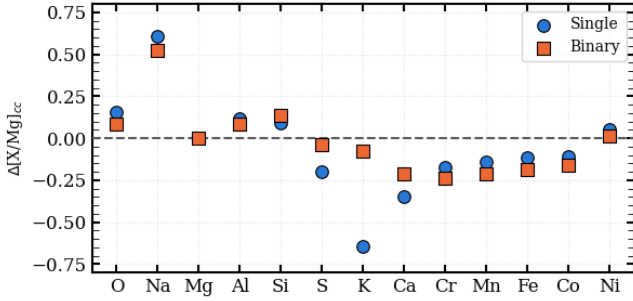


Figure 22. Offset between the theoretical CCSN abundance ratios from this work and empirical measurements of the CCSN contribution to $[X/Mg]$ for the Ms models (blue circles) and Mb models (red squares). Empirical constraints from Griffith et al. (2024). The dashed line at 0.0 represents agreement between the theoretical and empirical values.

with previous PUSH calculations (Ebinger et al. 2019; Curtis et al. 2019) and those of Farmer et al. (2023) and Boccioli & Roberti (2026). Our main conclusions are as follows:

(i) We find that binary stripping alters the explodability landscape across the mass range. The two series produce identical outcomes up to and including $22 M_{\odot}$, with all models successfully exploding, while their outcomes diverge at higher masses. Binary stripping can therefore alter the pre-collapse structure sufficiently to change the explosion outcome at a given ZAMS mass, enabling explosions in some models and preventing them in others. Overall, 18 of the 35 single-star progenitors and 18 of the 31 binary-stripped progenitors explode successfully, corresponding to success rates of 51.4 and 58.1 per cent, respectively. Despite this slightly higher success rate, the binary-stripped models are not systematically more explodable.

(ii) The predicted explodability landscape depends strongly on the adopted explosion treatment. Our PUSH calculations and those of Farmer et al. (2023) and Boccioli & Roberti (2026) agree on the outcomes of only 8 of the 66 models, with $11 M_{\odot}$ the only ZAMS mass at which all predict successful explosions for both the single-star and binary-stripped progenitors. Where the same models explode, their predicted neutron-star masses are broadly comparable. We find that our predicted outcomes are also sensitive to the adopted calibration, as changing the PUSH factor can be sufficient to reverse whether a model explodes. We therefore stress that our explodability landscape depends on the assumption that compactness is the appropriate quantity with which to calibrate the additional PUSH heating.

(iii) We find that the successfully exploding single-star and binary-stripped models occupy broadly similar regions of explosion parameter space, with energies of ~ 0.65 – 1.3 B, neutron-star masses of ~ 1.5 – $2.0 M_{\odot}$, and explosion times of ~ 0.35 – 0.65 s. The differences introduced by binary stripping are comparable to the scatter among the progenitor sets used in previous PUSH calculations (Ebinger et al. 2019), which were evolved using nuclear reaction networks of different sizes. We find that explosion properties show clearer trends with compactness than with ZAMS mass: more compact progenitors generally produce more massive neutron stars and explode later, while explosion energies peak at intermediate compactness and decrease towards both lower and higher values.

(iv) After nucleosynthesis post-processing of the successful explosions, we find that the largest systematic differences between the single-star and binary-stripped yields occur for the lightest elements, with single stars ejecting more material from their retained envelopes. Differences also remain when the yields are normalised by the total pre-explosion elemental yields or the Fe yield. At ZAMS masses

of ~ 11 – $14 M_{\odot}$, binary stripping substantially alters the yields of elements from S to Sc, which probe material processed through explosive O and Si burning. While we find that the Fe-group yields also exhibit scatter in the yields, this reflects differences in the structure close to the mass cut. At higher ZAMS masses, the elemental yields of the two progenitor series are generally more comparable.

(v) Using a Salpeter IMF weighting, we find that the single-star and binary-stripped populations produce broadly similar elemental yield patterns, with most elements differing by only ~ 20 per cent. The binary-stripped population shows moderate enhancements in Cl, Ar, and Sc, while N, F, Cu, and several Fe-group elements are suppressed. The clearest difference occurs for K, which is enhanced by a factor of ~ 3.4 , corresponding to an increase of ~ 0.63 dex in $[K/Fe]$. This enhancement reflects increased ^{39}K production in the lowest-mass progenitors, whose contributions receive greater weight under the Salpeter IMF, consistent with Farmer et al. (2023).

(vi) Our predicted ^{56}Ni yields span much of the observed CCSN range below $\sim 23 M_{\odot}$, and the iron-group abundance patterns broadly agree with those of HD 84937. The clearest discrepancy occurs for Cu, with predicted abundances above the observed value. However, Cu production is metallicity dependent, and our progenitors have solar metallicity whereas HD 84937 is very metal-poor, so some disagreement is not unexpected. The $[\alpha/Fe]$ ratios show the expected behaviour with ZAMS mass, with $^{16}\text{O}/Fe$ increasing most clearly and the Si and Ca ratios showing greater sensitivity to the explosive processing. Our IMF-averaged $[\alpha/Fe]$ ratios show broadly similar agreement with the observational constraints (Weinberg et al. 2024) for both populations, with the clearest difference occurring for Ca. For $^{40}\text{Ca}/Fe$, the single-star population lies substantially below the plateau by 0.37 dex, whereas the binary-stripped population shows a smaller deficit of 0.17 dex.

(vii) We find that binary stripping improves agreement with the empirical CCSN $[X/Mg]$ ratios of Griffith et al. (2024) for S, K, and Ca, while most other ratios differ by less than 0.1 dex between the two populations. Most elements lie within -0.25 to 0.15 dex of the empirical values. The largest discrepancy remains Na, with ratios above the empirical value by 0.5 dex for the binary-stripped population and 0.6 dex for the single-star population. The smaller excesses in O, Al, and Si may reflect an underproduction of Mg. Binary stripping therefore improves agreement for particular elements, but does not resolve all discrepancies between predicted and empirical CCSN abundance patterns.

(viii) We compare the IMF-averaged $[X/Mg]$ ratios of our single-star and binary-stripped progenitors with empirical CCSN abundance ratios inferred by Griffith et al. (2024) using a two-process model of SDSS DR17 APOGEE stellar abundances. Most elements lie between -0.25 and 0.15 dex of the empirical values. The largest discrepancy occurs for Na, with excesses of 0.5 dex for the binary-stripped population and 0.6 dex for the single-star population, while the smaller excesses in O, Al, and Si may reflect an underproduction of Mg. The two populations differ by less than 0.1 dex for most elements, with the clearest differences occurring for S, K, and Ca. Although both populations show similar abundance patterns overall, the closer agreement of the binary-stripped models with the empirical S, K, and Ca ratios highlights the importance of accounting for binary stripping when predicting the CCSN contribution to galactic chemical enrichment.

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DATA AVAILABILITY

The explosion properties and nucleosynthesis data are available from the corresponding author upon reasonable request. They will also be made publicly available on Zendo and via the Heidelberg Supernova Model Archive (HESMA; Kromer et al. 2017; <https://hesma.hits.org>).

REFERENCES

- Abomalima A., Frebel A., 2018, *ApJS*, **238**, 36
- Abt H. A., 1983, *ARA&A*, **21**, 343
- Aloy M. Á., Obergaulinger M., 2021, *MNRAS*, **500**, 4365
- Andrews S., Fryer C., Even W., Jones S., Pignatari M., 2020, *ApJ*, **890**, 35
- Andrievsky S., Bonifacio P., Caffau E., Korotin S., Spite M., Spite F., Sbordone L., Zhukova A. V., 2018, *MNRAS*, **473**, 3377
- Aoki W., et al., 2022, *ApJ*, **931**, 146
- Arentsen A., Aguado D. S., Sestito F., González Hernández J. I., Martín N. F., Starkenburg E., Jablonka P., Yuan Z., 2023, *MNRAS*, **519**, 5554
- Asplund M., Grevesse N., Sauval A. J., Scott P., 2009, *ARA&A*, **47**, 481
- Aufferheide M. B., Baron E., Thielemann F.-K., 1991, *ApJ*, **370**, 630
- Blondin J. M., Mezzacappa A., DeMarino C., 2003, *ApJ*, **584**, 971
- Boccioli L., Roberti L., 2026, *A&A*, **709**, A201
- Boccioli L., Mathews G. J., O'Connor E. P., 2021, *ApJ*, **912**, 29
- Braun H., Langer N., 1995a, in van der Hucht K. A., Williams P. M., eds, IAU Symposium Vol. 163, Wolf-Rayet Stars: Binaries; Colliding Winds; Evolution. p. 305
- Braun H., Langer N., 1995b, *A&A*, **297**, 483
- Brown G. E., Weingartner J. C., Wijers R. A. M. J., 1996, *ApJ*, **463**, 297
- Bruenn S. W., et al., 2016a, *ApJ*, **818**, 123
- Bruenn S. W., et al., 2016b, *ApJ*, **818**, 123
- Burrows A., 2013, *Reviews of Modern Physics*, **85**, 245
- Burrows A., Vartanyan D., Dolence J. C., Skinner M. A., Radice D., 2018, *Space Sci. Rev.*, **214**, 33
- Burrows A., Radice D., Vartanyan D., Nagakura H., Skinner M. A., Dolence J. C., 2020, *MNRAS*, **491**, 2715
- Burrows A., Wang T., Vartanyan D., 2024, *ApJ*, **964**, L16
- Burrows A., Wang T., Vartanyan D., 2026, *arXiv e-prints*, p. [arXiv:2607.06664](https://arxiv.org/abs/2607.06664)
- Caffau E., et al., 2013, *A&A*, **560**, A71
- Cayrel R., et al., 2004, *A&A*, **416**, 1117
- Couch S. M., Warren M. L., O'Connor E. P., 2020, *ApJ*, **890**, 127
- Curtis S., Ebinger K., Fröhlich C., Hempel M., Perego A., Liebendörfer M., Thielemann F.-K., 2019, *ApJ*, **870**, 2
- Curtis S., Wolfe N., Fröhlich C., Miller J. M., Wollaeger R., Ebinger K., 2021, *ApJ*, **921**, 143
- Cybert R. H., et al., 2010, *ApJS*, **189**, 240
- Da Costa G. S., et al., 2019, *MNRAS*, **489**, 5900
- Dessart L., 2025, *A&A*, **703**, A287
- Ebinger K., Curtis S., Fröhlich C., Hempel M., Perego A., Liebendörfer M., Thielemann F.-K., 2019, *ApJ*, **870**, 1
- Ebinger K., Curtis S., Ghosh S., Fröhlich C., Hempel M., Perego A., Liebendörfer M., Thielemann F.-K., 2020a, *ApJ*, **888**, 91
- Ebinger K., Curtis S., Ghosh S., Fröhlich C., Hempel M., Perego A., Liebendörfer M., Thielemann F.-K., 2020b, *ApJ*, **888**, 91
- Eldridge J. J., Izzard R. G., Tout C. A., 2008, *MNRAS*, **384**, 1109
- Ertl T., Janka H.-T., Woosley S. E., Sukhbold T., Ugliano M., 2016, *ApJ*, **818**, 124
- Farmer R., Fields C. E., Petermann I., Dessart L., Cantiello M., Paxton B., Timmes F. X., 2016, *ApJS*, **227**, 22
- Farmer R., Laplace E., de Mink S. E., Justham S., 2021, *ApJ*, **923**, 214
- Farmer R., Laplace E., Ma J.-z., de Mink S. E., Justham S., 2023, *ApJ*, **948**, 111
- Farouqi K., Kratz K.-L., Pfeiffer B., Rauscher T., Thielemann F.-K., Truran J. W., 2010, *ApJ*, **712**, 1359
- Fischer T., Martínez-Pinedo G., Hempel M., Liebendörfer M., 2012, *Phys. Rev. D*, **85**, 083003
- Foglizzo T., Galletti P., Scheck L., Janka H.-T., 2007, *ApJ*, **654**, 1006
- Fröhlich C., et al., 2006, *ApJ*, **637**, 415
- Grevesse N., Sauval A. J., 1998, *Space Sci. Rev.*, **85**, 161
- Griffith E. J., Sukhbold T., Weinberg D. H., Johnson J. A., Johnson J. W., Vincenzo F., 2021, *ApJ*, **921**, 73
- Griffith E. J., Johnson J. A., Weinberg D. H., Ilyin I., Johnson J. W., Rodriguez-Martinez R., Strassmeier K. G., 2023, *ApJ*, **944**, 47
- Griffith E. J., Hogg D. W., Dalcanton J. J., Hasselquist S., Ratcliffe B., Ness M., Weinberg D. H., 2024, *AJ*, **167**, 98
- Griffith E. J., Blum M., Weinberg D. H., Johnson J. A., Sit T., Ilyin I., Strassmeier K. G., 2026, *ApJ*, **1001**, 193
- Halevi G., Mösta P., 2018, *MNRAS*, **477**, 2366
- Hansen T., et al., 2015, *ApJ*, **807**, 173
- Harris C. R., et al., 2020, *Nature*, **585**, 357
- Heger A., Woosley S. E., 2002, *ApJ*, **567**, 532
- Heger A., Woosley S. E., Martínez-Pinedo G., Langanke K., 2001, *ApJ*, **560**, 307
- Heger A., Woosley S. E., Rauscher T., Hoffman R. D., Boyes M. M., 2002, *New Astron. Rev.*, **46**, 463
- Heger A., Kolbe E., Haxton W., Langanke K., Martínez-Pinedo G., Woosley S., 2005, *Physics Letters B*, **606**, 258
- Hempel M., Schaffner-Bielich J., 2010a, *Nuclear Physics A*, **837**, 210
- Hempel M., Schaffner-Bielich J., 2010b, *Nuclear Physics A*, **837**, 210
- Herant M., Benz W., Hix W. R., Fryer C. L., Colgate S. A., 1994, *ApJ*, **435**, 339
- Hirschi R., 2015, in Vink J. S., ed., *Astrophysics and Space Science Library Vol. 412, Very Massive Stars in the Local Universe*. p. 157 ([arXiv:1409.7053](https://arxiv.org/abs/1409.7053)), doi:10.1007/978-3-319-09596-7_6
- Hix W. R., et al., 2016, *Acta Physica Polonica B*, **47**, 645
- Howes L. M., et al., 2015, *Nature*, **527**, 484
- Hunter J. D., 2007, *Computing in Science & Engineering*, **9**, 90
- Jacobson H. R., et al., 2015, *ApJ*, **807**, 171
- Jalin G., Arcones A., O'Connor E. P., Ercolino A., Langer N., 2026, *arXiv e-prints*, p. [arXiv:2610.08617](https://arxiv.org/abs/2610.08617)
- Janka H.-T., 2012, *Annual Review of Nuclear and Particle Science*, **62**, 407
- Janka H.-T., 2025, *Annual Review of Nuclear and Particle Science*, **75**, 425
- Janka H.-T., Mueller E., 1996, *A&A*, **306**, 167
- Janka H.-T., Melson T., Summa A., 2016, *Annual Review of Nuclear and Particle Science*, **66**, 341
- Jerkstrand A., et al., 2015, *MNRAS*, **448**, 2482
- Jermyn A. S., et al., 2023, *ApJS*, **265**, 15
- Keller S. C., et al., 2014, *Nature*, **506**, 463
- Kromer M., Ohlmann S., Röpke F. K., 2017, *Mem. Soc. Astron. Italiana*, **88**, 312
- Kuroda T., Arcones A., Takiwaki T., Kotake K., 2020, *ApJ*, **896**, 102
- Langanke K., Martínez-Pinedo G., 2001, *Atomic Data and Nuclear Data Tables*, **79**, 1
- Langer N., 2012, *ARA&A*, **50**, 107
- Laplace E., Justham S., Renzo M., Göteborg Y., Farmer R., Vartanyan D., de Mink S. E., 2021, *A&A*, **656**, A58
- Lentz E. J., et al., 2015, *ApJ*, **807**, L31
- Li H.-N., Aoki W., Honda S., Zhao G., Christlieb N., Suda T., 2015a, *Research in Astronomy and Astrophysics*, **15**, 1264
- Li H., Aoki W., Zhao G., Honda S., Christlieb N., Suda T., 2015b, *PASJ*, **67**, 84
- Li H., et al., 2022, *ApJ*, **931**, 147
- Liebendörfer M., Mezzacappa A., Thielemann F.-K., 2001, *Phys. Rev. D*, **63**, 104003

Liebrandt M., Messer O. E. B., Mezzacappa A., Bruenn S. W., Cardall C. Y., Thielemann F.-K., 2004, *ApJS*, 150, 263

Limongi M., Chieffi A., 2006, *ApJ*, 647, 483

Limongi M., Chieffi A., 2012, *ApJS*, 199, 38

Limongi M., Chieffi A., 2018, *ApJS*, 237, 13

Lodders K., Palme H., Gail H.-P., 2009, *Landolt Börnstein*, 4B, 712

Majewski S. R., et al., 2017, *AJ*, 154, 94

Marino A. F., et al., 2019, *MNRAS*, 485, 5153

Martin N. F., et al., 2024, *A&A*, 692, A115

Mason B. D., Hartkopf W. I., Gies D. R., Henry T. J., Helsel J. W., 2009, *AJ*, 137, 3358

Melson T., Janka H.-T., Marek A., 2015a, *ApJ*, 801, L24

Melson T., Janka H.-T., Bollig R., Hanke F., Marek A., Müller B., 2015b, *ApJ*, 808, L42

Moe M., Di Stefano R., 2017, *ApJS*, 230, 15

Morozova V., Piro A. L., Renzo M., Ott C. D., Clausen D., Couch S. M., Ellis J., Roberts L. F., 2015, *ApJ*, 814, 63

Morrell N., Folatelli G., Phillips M., Contreras C., Hamuy M., 2005, *Central Bureau Electronic Telegrams*, 139, 1

Mösta P., et al., 2014, *ApJ*, 785, L29

Mösta P., Ott C. D., Radice D., Roberts L. F., Schnetter E., Haas R., 2015, *Nature*, 528, 376

Müller B., 2015, *MNRAS*, 453, 287

Nakamura K., Takiwaki T., Kuroda T., Kotake K., 2015, *PASJ*, 67, 107

Neopane S., Sandoval M. A., Hix W. R., Harris J. A., Messer O. E. B., Lentz E. J., 2026, *ApJ*, 1008, 36

Nishimura N., Takiwaki T., Thielemann F.-K., 2015, *ApJ*, 810, 109

Nishimura N., Sawai H., Takiwaki T., Yamada S., Thielemann F.-K., 2017, *ApJ*, 836, L21

Nomoto K., Kobayashi C., Tominaga N., 2013, *ARA&A*, 51, 457

Norris J. E., et al., 2013, *ApJ*, 762, 25

O'Connor E., Ott C. D., 2011, *ApJ*, 730, 70

Obergaulinger M., Aloy M. Á., 2020, *MNRAS*, 492, 4613

Paxton B., Bildsten L., Dotter A., Herwig F., Lesaffre P., Timmes F., 2011, *ApJS*, 192, 3

Paxton B., et al., 2013, *ApJS*, 208, 4

Paxton B., et al., 2015, *ApJS*, 220, 15

Paxton B., et al., 2018, *ApJS*, 234, 34

Paxton B., et al., 2019, *ApJS*, 243, 10

Perego A., Hempel M., Fröhlich C., Ebinger K., Eichler M., Casanova J., Liebrandt M., Thielemann F.-K., 2015, *Astrophys. J.*, 806, 275

Pignatari M., et al., 2016, *ApJS*, 225, 24

Placco V. M., Frebel A., Lee Y. S., Jacobson H. R., Beers T. C., Pena J. M., Chan C., Heger A., 2015, *ApJ*, 809, 136

Podsiadlowski P., Joss P. C., Hsu J. J. L., 1992, *ApJ*, 391, 246

Polis O. R., Tout C. A., Eggleton P. P., Han Z., 1995, *MNRAS*, 274, 964

Rauscher T., Thielemann F.-K., 2000, *Atomic Data and Nuclear Data Tables*, 75, 1

Rauscher T., Heger A., Hoffman R. D., Woosley S. E., 2002, *ApJ*, 576, 323

Rodríguez Ó., Maoz D., Nakar E., 2023, *ApJ*, 955, 71

Roederer I. U., Placco V. M., Beers T. C., 2016, *ApJ*, 824, L19

Rybicki J., Just A., Rix H.-W., 2017, *A&A*, 605, A59

Salpeter E. E., 1955, *ApJ*, 121, 161

Sana H., et al., 2012, *Science*, 337, 444

Sandoval M. A., Hix W. R., Messer O. E. B., Lentz E. J., Harris J. A., 2021, *ApJ*, 921, 113

Sawada R., Maeda K., 2019, *ApJ*, 886, 47

Sieverding A., Kresse D., Janka H.-T., 2023, *ApJ*, 957, L25

Snedden C., Cowan J. J., Kobayashi C., Pignatari M., Lawler J. E., Den Hartog E. A., Wood M. P., 2016, *ApJ*, 817, 53

Starkenburg E., et al., 2017, *MNRAS*, 471, 2587

Steiner A. W., Hempel M., Fischer T., 2013, *ApJ*, 774, 17

Sukhbold T., Ertl T., Woosley S. E., Brown J. M., Janka H.-T., 2016a, *ApJ*, 821, 38

Sukhbold T., Ertl T., Woosley S. E., Brown J. M., Janka H.-T., 2016b, *ApJ*, 821, 38

Takiwaki T., Kotake K., Suwa Y., 2014, *ApJ*, 786, 83

Thielemann F.-K., Nomoto K., Hashimoto M.-A., 1996, *ApJ*, 460, 408

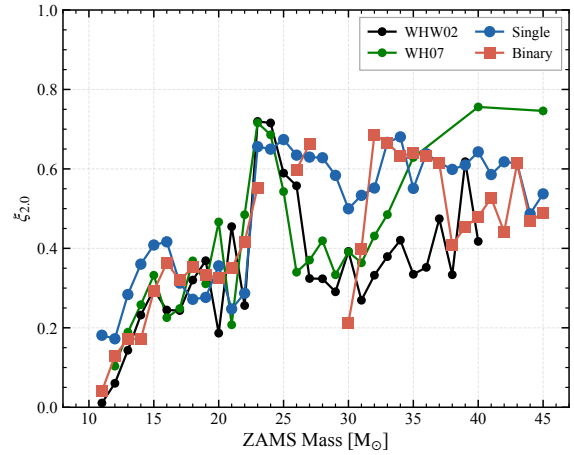


Figure A1. Similar to Figure 1 but for $\xi_{2.0}$ compactness

Timmes F. X., Swesty F. D., 2000, *ApJS*, 126, 501

Tout C. A., Polis O. R., Eggleton P. P., Han Z., 1996, *MNRAS*, 281, 257

Umeda H., Nomoto K., 2008, *ApJ*, 673, 1014

Vartanyan D., Burrows A., 2023, *MNRAS*, 526, 5900

Vartanyan D., Burrows A., Radice D., Skinner M. A., Dolence J., 2018, *MNRAS*, 477, 3091

Vartanyan D., Laplace E., Renzo M., Götzberg Y., Burrows A., de Mink S. E., 2021, *ApJ*, 916, L5

Venn K. A., et al., 2020, *MNRAS*, 492, 3241

Virtanen P., et al., 2020, *Nature Methods*, 17, 261

Weinberg D. H., Griffith E. J., Johnson J. W., Thompson T. A., 2024, *ApJ*, 973, 122

Wellstein S., Langer N., 1999, *A&A*, 350, 148

Winteler C., Käppeli R., Perego A., Arcones A., Vasset N., Nishimura N., Liebrandt M., Thielemann F.-K., 2012, *ApJ*, 750, L22

Wollaeger R. T., van Rossum D. R., Graziani C., Couch S. M., Jordan George C. I., Lamb D. Q., Moses G. A., 2013, *ApJS*, 209, 36

Woosley S., Heger A., 2007, *Physics Reports*, 442, 269

Woosley S. E., Weaver T. A., 1995, *ApJS*, 101, 181

Woosley S. E., Hartmann D. H., Hoffman R. D., Haxton W. C., 1990, *ApJ*, 356, 272

Woosley S. E., Langer N., Weaver T. A., 1993, *ApJ*, 411, 823

Woosley S. E., Heger A., Weaver T. A., 2002, *Reviews of Modern Physics*, 74, 1015

Yong D., et al., 2021, *MNRAS*, 507, 4102

Zepeda J., Rasmussen K. C., Beers T. C., Placco V. M., Huang Y., Depagne É., 2022, *ApJ*, 927, 13

de Mink S. E., Langer N., Izzard R. G., Sana H., de Koter A., 2013, *ApJ*, 764, 166

APPENDIX A: COMPACTNESS AS A FUNCTION OF ZAMS MASS

Figure A1 shows the compactness evaluated at $2.0 M_{\odot}$ at bounce, $\xi_{2.0}$, which is used to determine the additional PUSH heating through k_{push} , as a function of ZAMS mass (see Ebinger et al. 2019).

APPENDIX B: ISOTOPIC YIELDS OF THE MS18.00 AND MB18.00 MODELS

Table B1 presents the isotopic yields of the Ms18.00 and Mb18.00 models. These models are used throughout Section 4 to illustrate the differences in nucleosynthesis between the single-star and binary-stripped progenitors. The table includes isotopic yields through the end of the Ni isotopes for the Ms18.00 and Mb18.00 models. The remaining isotopic yields for these two models, together with the complete yield tables for all other successfully exploding models, are provided electronically.

Table B1: Pre-explosion, net change, and final isotopic yields, in units of $M_{\odot}$, for the representative Ms18.00 and Mb18.00 models. The net change is defined as the final isotopic mass minus the corresponding pre-supernova mass after all tracers have cooled below 1.75 GK. Negative and positive values therefore indicate a decrease or increase, respectively, in the total mass of a given isotope over the course of collapse and explosion. Note that these net changes can also reflect the reprocessing of material during collapse and explosion, such that substantial destruction and subsequent re-production of an isotope may result in only a small net change in its final abundance.

Isotope	Ms18			Mb18		
	Pre-explosion	Explosive	Final	Pre-explosion	Explosive	Final
^1H	1.76E+00	-4.71E-04	1.76E+00	7.28E-04	-5.52E-04	1.76E-04
^2H	9.64E-09	-9.64E-09	4.58E-17	1.57E-08	-1.57E-08	7.09E-20
^3He	5.16E-05	-2.98E-11	5.16E-05	5.22E-11	-5.08E-11	1.37E-12
^4He	2.67E+00	-2.11E-03	2.67E+00	1.28E+00	-3.66E-03	1.28E+00
^6Li		1.36E-19	1.36E-19		5.52E-20	5.52E-20
^7Li	2.41E-12	1.19E-11	1.43E-11	3.72E-12	3.32E-11	3.69E-11
^9Be	4.03E-11	-1.87E-12	3.84E-11	3.20E-12	-3.20E-12	1.66E-19
^{10}B	1.99E-10	-1.42E-13	1.99E-10	3.44E-13	-3.43E-13	1.65E-16
^{11}B	8.92E-10	6.01E-11	9.52E-10	1.59E-12	2.13E-11	2.29E-11
^{12}C	3.06E-01	7.05E-05	3.07E-01	2.46E-01	-6.36E-04	2.45E-01

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